τ AND ϵ INVARIANTS OF GENERALIZED MAZUR PATTERNS

RIBHU HOOJA, SEAN KIM, FILIP RUPCHIN, SHREYA SINHA, AND ALLISON TSYPIN

Abstract. We generalize work by Patwardhan and Xiao for a three-parameter infinite family of pattern knots $Q_{i,(m,n)}$ for which we derive closed formulas for the τ and ϵ concordance invariants of the satellite knot $Q_{i,(m,n)}(K)$ for any companion knot K. We rely on immersed curve techniques to compute bordered Heegaard Floer homology arising from the work of Chen-Hanselman, and we provide an alternative computation for a one-parameter infinite subfamily using more traditional bordered Heegaard Floer techniques.

1. Introduction

Two knots in S^3 are *smoothly concordant* if they co-bound a smoothly embedded annulus in $S^3 \times I$. A knot $K \subset S^3$ is *smoothly slice* if it is smoothly concordant to the unknot. The equivalence classes of knots classified up to smooth concordance form the *smooth knot concordance group*, denoted $\mathcal{C}$, with connect-sum inducing the group operation. Two knots in S^3 are *rationally concordant* if they co-bound a smoothly embedded annulus in a compact, smooth 4-manifold W with $\partial W = -S^3 \sqcup S^3$ and $H_*(W;\mathbb{Q}) \cong H_*(S^3 \times I;\mathbb{Q})$. A knot $K \subset S^3$ that is rationally concordant to the unknot is *rationally slice*. Using this notion of concordance defines the *rational homology concordance group* $\mathcal{C}_\mathbb{Q}$.

Much of the structure of the knot concordance groups remains unknown. One approach to studying $\mathcal{C}$ and $\mathcal{C}_\mathbb{Q}$ is satellite operators. A *pattern knot* P is a knot $P \subset S^1 \times D^2$ that assigns to every $K \subset S^3$ (called a *companion knot*) a *satellite knot* $P(K) \subset S^3$ defined as the image of P under the canonical identification of a tubular neighborhood of K denoted $\nu(K)$ with the solid torus $S^1 \times D^2$. Satellite operations respect knot concordances; thus they induce well-defined maps on $\mathcal{C}$ and $\mathcal{C}_\mathbb{Q}$ [Lev21]; however, it is conjectured that these maps are not generally homomorphisms [HPC21].

Nonetheless, satellite operators remain a useful tool in the study of knot concordance. Using the Mazur pattern Q, displayed in Figure 1, Levine answers the following question in the affirmative [Lev21] by showing that Q induces a non-surjective satellite operator on $\mathcal{C}_\mathbb{Q}$:

Problem 1.1. *Does there exist a winding number ± 1 pattern knot P such that $P(K)$ is not rationally slice for every knot $K \subset S^3$?*

Levine proves that the Mazur pattern induces a non-surjective satellite operator by computing closed formulas for the τ [OS03] and ϵ [Hom14] knot invariants of Mazur satellite knots using bordered Heegaard Floer homology. The τ invariant is a group homomorphism $\tau : \mathcal{C}_\mathbb{Q} \to \mathbb{Z}$ and $\epsilon : \mathcal{C}_\mathbb{Q} \to \{-1, 0, 1\}$ is a well-defined map, whose behavior under connect sum mimics that of the sign function on integers under addition. In particular, if K is rationally slice then $\tau(K) = \epsilon(K) = 0$. Both τ and ϵ are rational homology concordance invariants: for τ this follows from the genus bound of Ozsváth and Szabó [OS03], and for ϵ it follows from Hom's cabling formula [Hom14]; see [Lev21, Section 1].

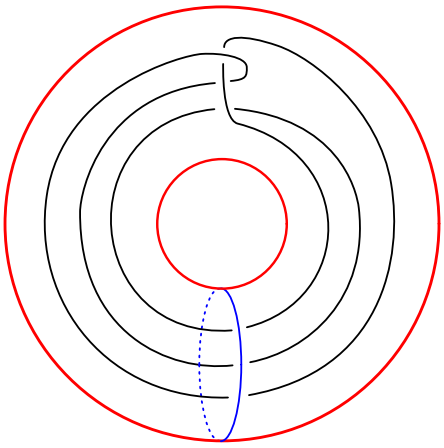


FIGURE 1. The Mazur pattern knot Q in the solid torus.

Recent work by Patwardhan–Xiao [PX24], and Bodish [Bod24] extends these computations to infinite families of generalized Mazur pattern knots using immersed curve techniques for bordered Heegaard Floer homology [HRW23]. In this paper, we extend these results using similar techniques to a three-parameter infinite family of generalized Mazur pattern knots denoted $Q_{i,(m,n)} \subset S^1 \times D^2$ with $i, m, n \geqslant 1$ as shown in Figure 2.

Theorem 1.2. *For $i, m, n \geqslant 1$, let $Q_{i,(m,n)}$ be a generalized Mazur pattern embedded in $S^1 \times D^2$, and consider any companion knot $K \subset S^3$. If $m \neq n$, we have*

$$\tau\left(Q_{i,(m,n)}(K)\right) = \begin{cases} |m-n|\tau(K) & \text{if } \tau(K) \leqslant 0 \text{ and } \epsilon(K) \in \{0,1\} \\ |m-n|\tau(K) + |m-n| & \text{if } \tau(K) < 0 \text{ and } \epsilon(K) = -1 \\ |m-n|\tau(K) + \min\{m,n\} & \text{if } \tau(K) > 0 \text{ and } \epsilon(K) = 1 \\ |m-n|\tau(K) + \max\{m,n\} - 1 & \text{if } \tau(K) \geqslant 0 \text{ and } \epsilon(K) = -1 \end{cases} \tag{1}$$

and

$$\epsilon\left(Q_{i,(m,n)}(K)\right) = \begin{cases} 0 & \text{if } \tau(K) = \epsilon(K) = 0 \\ 1 & \text{otherwise.} \end{cases} \tag{2}$$

When $m = n$, we have

$$\tau\left(Q_{i,(m,m)}(K)\right) = \begin{cases} 0 & \text{if } \tau(K) < 0, \text{ or } \tau(K) = 0 \text{ and } \epsilon(K) \in \{0,1\} \\ m-1 & \text{if } \tau(K) = 0 \text{ and } \epsilon(K) = -1 \\ m & \text{if } \tau(K) > 0 \end{cases} \tag{3}$$

and

$$\epsilon\left(Q_{i,(m,m)}(K)\right) = \begin{cases} 0 & \text{if } \tau(K) = 0 \text{ and } \epsilon(K) \in \{0,1\} \\ 1 & \text{otherwise.} \end{cases} \tag{4}$$

Remark 1.3. Patwardhan and Xiao's [PX24] τ and ϵ calculations correspond to the family $Q_{1,(m,n)}$ with $m, n \geqslant 1$ in our notation. Their results match our computations with the exception of the ϵ invariant for the $m = n$ case, where they found $\epsilon(Q_{1,(m,n)}(K)) = 0$ when $\tau(K) = 0$ and $\epsilon(K) = 1$.

As an immediate corollary we find a new two-parameter family of knots resolving Problem 1.45 [Kir97].

Corollary 1.4. *Let $T_{2,3}$ denote the right-handed trefoil. For any $i \geqslant 1$ and $m \geqslant 2$, the family of winding number ± 1 pattern knots $P_{i,m} := Q_{i,(m,m-1)} \# T_{2,3} \subset S^1 \times D^2$ is such that for any knot $K \subset S^3$, $P_{i,m}(K)$ is not rational homology slice.*

Proof. Let $K \subset S^3$ be a knot. Theorem 1.2 implies that $\epsilon(Q_{i,(m,m-1)}(K)) \in \{0, 1\}$. Using that $\epsilon(T_{2,3}) = 1$, the behavior of $\epsilon : \mathcal{C}_{\mathbb{Q}} \to \{-1, 0, 1\}$ under connect sum tells us:

$$\epsilon(P_{i,m}(K)) = \epsilon((Q_{i,(m,m-1)} \# T_{2,3})(K)) = \epsilon(Q_{i,(m,m-1)}(K) \# T_{2,3}) = 1.$$

The conclusion follows from the observation that the winding number $w(Q_{i,(m,m-1)}) = 1$. □

Furthermore, Theorem 1.2 represents a significant step towards the resolution of a question posed by Patwardhan and Xiao [PX24].

Question 1.5 (Patwardhan and Xiao [PX24])**.** *Is there a closed formula for τ of satellite knots with $(1, 1)$-unknot patterns?*

In [Che23], Chen establishes a one-to-one correspondence between (1,1)-unknot patterns and two-bridge links. We show that the generalized Mazur pattern family is the natural extension of Patwardhan and Xiao's generalized Mazur when drawn in two-bridge link form. Thus, we hope that our computations and techniques can serve as a solid foundation for future work attempting to resolve Question 1.5, in addition to a possible closed formula for ϵ of satellite knots with $(1, 1)$-unknot patterns.

Question 1.6. *Is there a closed formula for ϵ of satellite knots with $(1, 1)$-unknot patterns?*

1.1. **Drawing generalized Mazur pattern knots.** Our generalized Mazur pattern family contains Patwardhan and Xiao's generalizations and partially overlaps with Bodish's.

Definition 1.7 (generalized Mazur patterns)**.** Fix $i, m, n \geqslant 1$. Let $\alpha \subset S^1 \times D^2$ be an arc that winds m times around the S^1 direction, reverses direction, and then winds n times around in the opposite direction. Complete α to a knot with an arc β that:

(1) passes under the n strands of the second pass of α,
(2) forms a twist region with the turn-around arc of α, consisting of i full negative twists, and
(3) passes over the m strands of the first pass of α.

The resulting knot is the *generalized Mazur pattern* $Q_{i,(m,n)} \subset S^1 \times D^2$. Note that $|w(Q_{i,(m,n)})| = |m - n|$. See Figure 2.

Remark 1.8. Bodish generalizes the standard Mazur pattern differently through the additions of $i - 1$ clasps to the standard Mazur pattern and n full meridional twists. We only incorporated Bodish's addition of clasps to our generalized Mazur pattern family. Using Bodish's notation $Q_n^{i,j}$ as described in Figure 3, our families overlap for $i, m \geqslant 1$ with $n = 1$ as $Q_{i,(m,1)} = Q_0^{i-1,m-1}$ [Bod24].

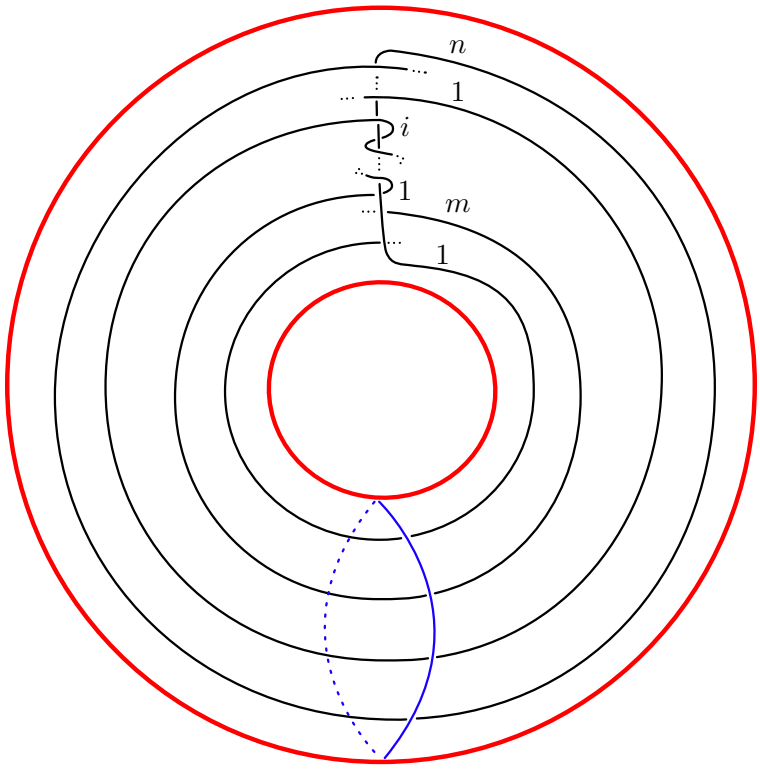


FIGURE 2. The generalized Mazur pattern knot $Q_{i,(m,n)}$ in the solid torus.

Structure of the paper. We first introduce the relevant definitions from Heegaard Floer homology in Section 2, knot Floer homology in Section 3, and bordered Heegaard Floer homology in Section 4. We derive the bordered Heegaard diagrams for generalized Mazur patterns in Section 5. We then use immersed curve techniques to prove Theorem 1.2 in Section 6, and then provide an alternative method for the computation of τ for satellites of $Q_{i,(2,2)}$ for $i \geqslant 1$ recovering the structure of $CFA^-(Q_{i,(2,2)})$ in Section 7.

Acknowledgments. The authors are grateful to Ina Petkova and Abhishek Mallick for their guidance, to Robert Lipshitz, Jacob Rasmussen, Ciprian Manolescu, and Michael Wong for helpful discussions, and to the Summer 2026 SHUR program for providing the opportunity for their research. The Summer 2026 SHUR program was partially supported by the Jack Byrne Scholarship program and NSF CAREER Grant DMS-2145090.

2. Heegaard Floer homology

Defined by Ozsváth and Szabó [OS04d, OS04c], Heegaard Floer homology is a variant of Lagrangian Floer homology associating homology groups to any closed oriented three-manifold Y whose chain complex has boundary maps defined by counting pseudo-holomorphic discs in the

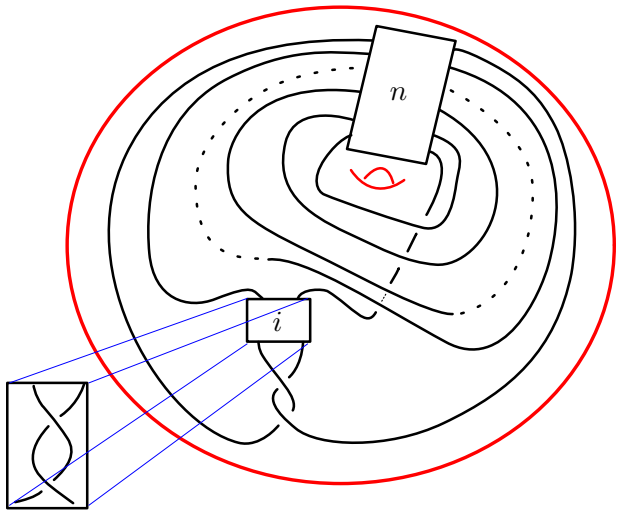


FIGURE 3. Bodish's generalized Mazur pattern knot $Q_n^{i,j}$. The box labelled i contains i full twists on two strands, as shown in the box on the bottom left. The box labelled n contains n full twists on $j+2$ strands.

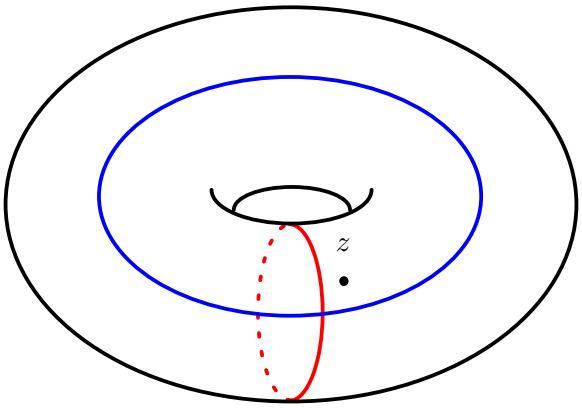


FIGURE 4. A genus-one Heegaard diagram for S^3. As is convention, the α-curve is drawn in red and the β-curve in blue.

symmetric product space of a Heegaard surface. The simplest version, denoted $\widehat{\mathrm{HF}}(Y)$, is a vector space over the field $\mathbb{F}$ with two elements. We will also discuss its other variant $\mathrm{HF}^-(Y)$. For a more detailed introduction to Heegaard Floer homology, see [OS06].

2.1. **Heegaard splittings and diagrams.** Given a closed oriented three-manifold Y, there exists a *Heegaard splitting* $Y = H_0 \cup_{\Sigma_g} H_1$, where H_0 and H_1 are handlebodies of the same genus g glued along their common boundary surface Σ_g [Moi52]. The genus g attaching surface of our diagram

Σ_g is called the *Heegaard surface*. For the purposes of this paper we will be working with the genus one Heegaard splitting of S^3 into two solid tori glued along a genus one Heegaard surface, i.e. a torus. This can easily be seen by taking a tubular neighborhood of the unknot in S^3 as one of the tori, and since the complement is also a solid torus, we get a genus one Heegaard decomposition of S^3.

We can specify a choice of Heegaard splitting by a *Heegaard diagram* consisting of a triple $H = (\Sigma_g, \alpha, \beta)$, where Σ_g is an oriented surface, $\alpha = \{\alpha_1, \ldots, \alpha_g\}$ and $\beta = \{\beta_1, \ldots, \beta_g\}$ are two complete sets of attaching circles for Σ_g. Here a *complete set of attaching circles* for a handlebody means a collection of g pairwise disjoint simple closed curves in Σ_g that are linearly independent in $H_1(\Sigma_g; \mathbb{Z})$ and each bound a compressing disc in that handlebody, so that the α curves specify H_0 and the β curves specify H_1. For technical reasons, when defining Heegaard Floer homology we add a basepoint w to the Heegaard diagram disjoint from the α and β circles. Three moves can be applied on Heegaard diagrams that do not change the underlying three-manifold: isotopies, handle slides and stabilizations, and any two Heegaard diagrams for the same 3-manifold can be connected by these moves. Thus, to define a 3-manifold invariant, it suffices to define it on the level of Heegaard diagrams, and prove invariance under these three types of moves. We include an example of a genus one pointed Heegaard diagram in Figure 4. For more details regarding the existence of Heegaard splittings and Heegaard diagrams please refer to sections 2 and 3 of [OS06].

2.2. **Symmetric products and Whitney discs.** Given any pointed Heegaard diagram $H = (\Sigma_g, \alpha, \beta, w)$ we define the *symmetric product* of the Heegaard surface:

$$\mathrm{Sym}^g(\Sigma_g) := \Sigma_g \times \ldots \times \Sigma_g \,/\, S_g$$

where S_g denotes the symmetric group on g letters. $\mathrm{Sym}^g(\Sigma_g)$ can be thought of as the set of all unordered g-tuples of points in Σ_g with repetition. Importantly, it can be shown that $\mathrm{Sym}^g(\Sigma_g)$ is a $2g$ real dimensional smooth manifold, and that an almost-complex structure on Σ_g induces one on $\mathrm{Sym}^g(\Sigma_g)$. Furthermore, the two sets of attaching circles α and β induce a pair of smoothly embedded g-dimensional tori in $\mathrm{Sym}^g(\Sigma_g)$,

$$\mathbb{T}_\alpha = \alpha_1 \times \ldots \times \alpha_g \quad \text{and} \quad \mathbb{T}_\beta = \beta_1 \times \ldots \times \beta_g.$$

Before proceeding, note that in the genus one case $\mathrm{Sym}^1(\Sigma_1) = \Sigma_1$ so the tori $\mathbb{T}_\alpha$ and $\mathbb{T}_\beta$ are just the two α and β circles.

Definition 2.1. Let D be the unit disc in $\mathbb{C}$. Let e_1, e_2 be the arcs in the boundary of D with $\Re(\cdot) \geqslant 0, \Re(\cdot) \leqslant 0$ respectively. Then given a pair of intersection points $x, y \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta$, a *Whitney disc* connecting x and y is a continuous map,

$$u : D \to \mathrm{Sym}^g(\Sigma_g)$$

with the properties that $u(-i) = x, u(i) = y, u(e_1) \subset \mathbb{T}_\alpha, u(e_2) \subset \mathbb{T}_\beta$. Let $\pi_2(x, y)$ denote the set of homotopy classes of Whitney discs connecting x and y.

The general theory of symmetric products of Heegaard surfaces and Whitney discs requires more careful treatment; however, the genus one case allows us to restrict our focus. For more details regarding the general theory of symmetric product spaces, Whitney discs, and Spin^c structures refer to sections 4, 5, and 6 respectively of [OS06].

2.3. **Holomorphic discs and Floer chain complexes.** An almost-complex structure on Σ_g induces an almost-complex structure on $\mathrm{Sym}^g(\Sigma_g)$. For a given homotopy class $\phi \in \pi_2(x, y)$, we define $\mathcal{M}(\phi)$ to be the moduli space of pseudoholomorphic representatives of ϕ. It can be shown that $\mathcal{M}(\phi)$ admits an $\mathbb{R}$-action corresponding to the complex automorphisms of the unit disc that preserve i and $-i$. Taking the quotient of $\mathcal{M}(\phi)$ by this $\mathbb{R}$ action, we define the moduli space:

$$\widehat{\mathcal{M}}(\phi) := \mathcal{M}(\phi)/\mathbb{R}.$$

Remark 2.2. By the Riemann mapping theorem, there exists a biholomorphism between $D\backslash\{\pm i\}$ and $[-1, 1] \times i\mathbb{R}$. The $\mathbb{R}$-action can be seen on this picture as translation in the $i\mathbb{R}$ direction.

To each homotopy class $\phi \in \pi_2(x, y)$ we can associate an integer $\mu(\phi)$, the *Maslov index*, which is the expected dimension of $\mathcal{M}(\phi)$ [RS93] for a generically chosen almost-complex structure on $\mathrm{Sym}^g(\Sigma_g)$. Since $\widehat{\mathcal{M}}(\phi)$ is the quotient of $\mathcal{M}(\phi)$ by a free $\mathbb{R}$ action, its expected dimension is $\mu(\phi) - 1$. In particular, when $\mu(\phi) = 1$ and the almost complex structure is chosen generically, it can be shown using the Gromov compactness theorem that $\widehat{\mathcal{M}}(\phi)$ is a compact zero dimensional manifold, hence a finite set of points, and we write $\#\widehat{\mathcal{M}}(\phi)$ for the number of these points (counted modulo two since we work over $\mathbb{F}$) [OS04c].

Remark 2.3. In the genus one case, $\mathrm{Sym}^1(\Sigma_1) = \Sigma_1$, so a Whitney disc is simply a map from D into the Riemann surface Σ_1. In particular we may work with honestly holomorphic discs rather than the pseudo-holomorphic discs required in the general theory. Moreover, we can decompose Σ_1 into *regions*, the connected components of $\Sigma_1\backslash(\alpha \cup \beta)$. A holomorphic disc covers each region a non-negative number of times, and these multiplicities depend only on the homotopy class of the disc; the resulting formal sum of regions is called the *domain* of ϕ. Combining this non-negativity with the index condition, the classes ϕ with $\mu(\phi) = 1$ that admit a holomorphic representative are precisely those whose domain is an embedded bigon in Σ_1, that is, a disc bounded by an arc of α and an arc of β meeting at the two corners x and y. Best of all, the Riemann mapping theorem gives each such bigon a unique holomorphic representative up to reparameterization by the $\mathbb{R}$ action described above, so $\#\widehat{\mathcal{M}}(\phi) = 1$.

In the general theory one must fix a Spin^c structure on Y, and the generators are sorted into equivalence classes according to the Spin^c structure they induce [OS06]. Thankfully, S^3 admits a unique Spin^c structure thus, no such sorting occurs and we may work with all of $\alpha \cap \beta$ at once.

The generators $x, y \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta$ can be endowed with a relative grading, called the *Maslov grading*. The difference between the Maslov gradings of any two generators is given by

$$M(x) - M(y) = \mu(\phi) - 2n_w(\phi),$$

where ϕ is any element of $\pi_2(x, y)$. That this is independent of the choice of ϕ follows from Proposition 5.14 and Lemma 7.1 of [OS06].

Definition 2.4. Let $H = (\Sigma_1, \alpha, \beta, w)$ be a pointed genus one Heegaard diagram. Then $\widehat{\mathrm{CF}}(H)$ is the $\mathbb{F}$-vector space generated by the intersection points $\alpha \cap \beta$, with differential:

$$\hat{\partial} x \;=\; \sum_{y \in \alpha \cap \beta} \; \sum_{\substack{\phi \in \pi_2(x,y) \\ \mu(\phi)=1,\, n_w(\phi)=0}} \#\widehat{\mathcal{M}}(\phi) \cdot y,$$

where $n_w(\phi)$ denotes the multiplicity of the region containing the basepoint for a disc ϕ.

Keeping track of the bigons that run over w, rather than discarding them, gives the minus variant of the Floer chain complex. In general one must require the diagram to satisfy a technical *admissibility* condition to ensure the sum below is finite [OS04d]; in the genus one case this holds automatically.

Definition 2.5. Let $H = (\Sigma_1, \alpha, \beta, w)$ be a pointed genus one Heegaard diagram. Then $\mathrm{CF}^-(H)$ is the free $\mathbb{F}[U]$-module generated by the intersection points $\alpha \cap \beta$, with differential

$$\partial^- x \;=\; \sum_{y \in \alpha \cap \beta} \sum_{\substack{\phi \in \pi_2(x,y) \\ \mu(\phi)=1}} \#\widehat{\mathcal{M}}(\phi)\, U^{n_w(\phi)} \cdot y,$$

where $n_w(\phi)$ denotes the multiplicity of the region containing the basepoint for a disc ϕ.

In both the $\widehat{\mathrm{CF}}$ and CF^- variants, $\partial^2 = 0$ and the resulting homologies $\widehat{\mathrm{HF}}(Y)$ and $\mathrm{HF}^-(Y)$ do not depend on the choices made, such as the Heegaard diagram or choice of almost-complex structure on Σ_g [OS04d]. Additionally, for both the $\widehat{\mathrm{CF}}$ and CF^- variant, ∂ drops the grading by one and U drops it by two. Furthermore, it is easy to see that:

$$\widehat{\mathrm{CF}}(H) \cong \mathrm{CF}^-(H)/U.$$

As an example, for the genus one diagram of S^3 with a single intersection point there is one generator and no differential, so $\widehat{\mathrm{HF}}(S^3) = \mathbb{F}$ and $\mathrm{HF}^-(S^3) = \mathbb{F}[U]$. For information regarding other variants of Heegaard Floer homology such as $\mathrm{HF}^\infty(Y)$ and $\mathrm{HF}^+(Y)$, please refer to section 8 of [OS06].

3. Knot Floer Homology

Defined by Ozsváth, Szabó [OS04b], and independently by Rasmussen [Ras03], knot Floer homology is a knot invariant based off of Heegaard Floer homology. Specifically, it associates to each knot a bi-graded finitely generated abelian group. Similarly to Heegaard Floer homology, knot Floer homology comes in different flavors, and we will present both the $\widehat{\mathrm{HFK}}$ and HFK^- variants. Furthermore, we will briefly present τ [OS04b, Ras03] and ϵ [Hom14]. In this section we restrict ourselves to the study of knots in S^3. For a more detailed overview of knot Floer homology please refer to [Man16].

3.1. Knot Heegaard diagrams.

Every knot in S^3 can be encoded combinatorially by a multi-pointed Heegaard diagram $(\Sigma_g, \boldsymbol{\alpha}, \boldsymbol{\beta}, z, w)$: a Heegaard surface Σ_g splitting S^3 into handlebodies H_0 and H_1, $g + k - 1$ attaching curves (α and β) cutting each handlebody into k balls, and two sets of k basepoints z and w recording where the knot crosses Σ_g. Such a diagram is called a genus g bridge k Heegaard diagram, or a (g, k)-diagram. Importantly, it can be shown that any knot K admits a genus g bridge one presentation, more often denoted a $(g, 1)$ presentation [Man16]. A knot admitting a $(1, 1)$ presentation is called a $(1, 1)$ knot.

We restrict ourselves to this case, where the Heegaard diagram becomes a standard *doubly pointed Heegaard diagram* $(\Sigma_g, \boldsymbol{\alpha}, \boldsymbol{\beta}, z, w)$. The knot can be recovered from the diagram by drawing two arcs: one in each handlebody H_i. The arc in H_0 is drawn without intersecting any of

the α curves and the arc in H_1 is drawn without intersecting any of the β curves. Similarly to singly-pointed Heegaard diagrams, doubly-pointed Heegaard diagrams produce the same knot Floer homologies up to isotopies, handleslides, and stabilizations, although now isotopies also cannot go over the z basepoint.

3.2. **Knot Floer complexes & homologies.** Constructions of the knot Floer complex are similar to those of standard Heegaard Floer complexes, with the difference that the set of generators $\mathbb{T}_\alpha \cap \mathbb{T}_\beta$ now carries two gradings. Throughout this section $H = (\Sigma, \alpha, \beta, w, z)$ is a doubly-pointed Heegaard diagram for $K \subset S^3$. The relative Maslov grading is identical to that of Heegaard Floer homology with,

$$M(x) - M(y) = \mu(\phi) - 2n_w(\phi), \qquad \phi \in \pi_2(x, y).$$

The second grading is the integer valued relative Alexander grading, defined by:

$$A(x) - A(y) = n_z(\phi) - n_w(\phi), \qquad \phi \in \pi_2(x, y).$$

Similarly to the relative Maslov grading, the Alexander grading is also independent of our choice of ϕ [OS17], so the relative Alexander gradings are well defined. We fix the absolute grading for A by requiring

$$\sum_{x \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta} (-1)^{M(x)} q^{A(x)} = \Delta_K(q),$$

where $\Delta_K(q)$ is the symmetrized Alexander polynomial of K, normalized so that $\Delta_K(q) = \Delta_K(q^{-1})$. The symmetry of Δ_K lets us determine the absolute grading.

Before proceeding, the filtered chain homotopy type of a knot Floer chain complex $C(H)$ is a knot invariant and we thus denote it by $C(K)$.

3.2.1. $\widehat{CFK}$. The simplest of the knot Floer complexes is denoted $\widehat{CFK}(K)$, the $\mathbb{F}$-vector space generated by $\mathbb{T}_\alpha \cap \mathbb{T}_\beta$ with differential counting only discs whose domain avoids w:

$$\partial x = \sum_{y \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta} \sum_{\substack{\phi \in \pi_2(x,y) \\ \mu(\phi)=1,\ n_w(\phi)=0}} \#\widehat{\mathcal{M}}(\phi) \cdot y.$$

If ϕ contributes to ∂ then $n_w(\phi) = 0$ and $n_z(\phi) \geqslant 0$, so $A(x) - A(y) = n_z(\phi) \geqslant 0$, thus $A(x) = A(y) + n_z(\phi)$ giving,

$$A(\partial x) \leqslant A(x),$$

where the left hand side denotes the maximum Alexander grading of the terms appearing in ∂x [Hom17]. The differential therefore never raises Alexander grading, and A induces what we call the *Alexander filtration* of $\widehat{CFK}(K)$. The Alexander filtration, rather than the homology of $\widehat{CFK}(K)$, carries the knot theoretic information. Specifically, for any $K \subset S^3$ we have that

$$H_*(\widehat{CFK}(K)) \cong \widehat{\mathrm{HF}}(S^3) \cong \mathbb{F},$$

supported in a single Maslov grading. By requiring that Maslov grading to zero, we can upgrade the relative Maslov grading to an absolute grading.

The associated graded complex $g\widehat{CFK}(K)$ is obtained by discarding the differentials that strictly drop Alexander grading, that is, by imposing the condition that $n_z(\phi) = 0$ as well:

$$\partial x \;=\; \sum_{y\in\mathbb{T}_\alpha\cap\mathbb{T}_\beta} \;\sum_{\substack{\phi\in\pi_2(x,y)\\ \mu(\phi)=1,\ n_w(\phi)=n_z(\phi)=0}} \#\widehat{\mathcal{M}}(\phi)\cdot y.$$

Its homology is the bigraded vector space denoted as,

$$\widehat{\mathrm{HFK}}(K) \;=\; \bigoplus_{i,s\in\mathbb{Z}} \widehat{\mathrm{HFK}}_i(K,s),$$

where i is the Maslov grading and s the Alexander grading.

3.2.2. *CFK^-.* The complex $CFK^-(K)$ is defined similarly to CF^-, as the free $\mathbb{F}[U]$-module generated by $\mathbb{T}_\alpha \cap \mathbb{T}_\beta$ with differential:

$$\partial^- x \;=\; \sum_{y\in\mathbb{T}_\alpha\cap\mathbb{T}_\beta} \;\sum_{\substack{\phi\in\pi_2(x,y)\\ \mu(\phi)=1}} \#\widehat{\mathcal{M}}(\phi)\, U^{n_w(\phi)}\cdot y.$$

Here domains of discs are permitted to contain the w basepoint, with multiplicity recorded by the exponent of U. We note that the admissibility conditions for H guarantee that the sum is finite. The differential drops Maslov grading by one and U drops it by two, while for the Alexander grading we have:

$$A(\partial^- x) \leqslant A(x) \quad \text{and} \quad A(Ux) = A(x) - 1,$$

Thus, we have again that A induces a filtration. Setting $U = 0$ recovers $\widehat{CFK}$:

$$\widehat{CFK}(K) \cong CFK^-(K)/U.$$

As before, $gCFK^-(K)$ denotes the associated graded complex, obtained by retaining only the differentials with $n_z(\phi) = 0$. Its homology $\mathrm{HFK}^-(K)$ is a bigraded module over $\mathbb{F}[U]$.

3.2.3. *CFK^∞.* We now define the *full knot Floer complex* CFK^∞: a freely generated $\mathbb{F}[U, U^{-1}]$ module over the generators $\mathbb{T}_\alpha \cap \mathbb{T}_\beta$, with the differential given by the same formula as that of CFK^-. We now describe a graphical encoding of CFK^∞.

More specifically, we can encode the full knot Floer complex graphically on a $\mathbb{Z}^2$ lattice by thinking of each generator x as a point with coordinates (i, j) with j representing the Alexander grading of $U^{-i}x$ and i as the negative of the U power. Following this convention, we then draw the differentials by arrows as follows. If a disc $\phi \in \pi_2(x, y)$ contributes to the differential, the change in horizontal coordinate is $-n_w(\phi)$ and the change in vertical coordinate is $-n_z(\phi)$. Since both $n_w(\phi)$ and $n_z(\phi)$ are nonnegative, every arrow points weakly left and weakly down.

We call a complex *reduced* if every arrow strictly decreases at least one of the two coordinates, so that no arrow is a self-map of a filtration level. Any $CFK^\infty(K)$ complex can be brought into reduced form by a *filtered change of basis* [Hom14]. We assume all our complexes are reduced from here on, since this is what allows the vertical and horizontal differentials of Section 3.3 to be read off directly from the arrows.

Note that we can recover the following from CFK^∞:

- If we only consider the points in CFK^∞ whose i-coordinate satisfies $i \geqslant 0$, we recover CFK^-.
- The quotient $CFK^+ := CFK^\infty/CFK^-$ can be used to recover $\widehat{\mathrm{CF}}(S^3)$ with the knot filtration given by the j-coordinate by only considering the generators with i-coordinate 0.
- If we ignore the vertical differentials and only consider the horizontal differentials in CFK^-, we obtain $gCFK^-$.

3.2.4. *Geometric Applications.* We present relevant geometric applications of $\widehat{\mathrm{HFK}}$. First, recall the definition of the *genus of a knot*:

$$g(K) = \min\{\text{genus}(F) \mid F \subset S^3 \text{ is an oriented, embedded surface with } \partial F = K\}.$$

In fact, a nice property of the Alexander polynomial is that its degree gives a lower bound on the genus [Sei35]. Specifically, if:

$$\Delta K(q) = a_0 + a_1(q + q^{-1}) + \ldots + a_n(q^n + q^{-n}), \quad a_n \neq 0,$$

then $g(K) \geqslant n$. Ozsváth and Szabó show that knot Floer homology strengthens this, showing that it directly detects the knot genus [OS04a].

Theorem 3.1. *[Ozsváth-Szabó, Theorem 1.2 in* [OS04a]*] For any knot $K \subset S^3$,*

$$g(K) = \max\{s \geqslant 0 \mid \widehat{\mathrm{HFK}}_*(K, s) \neq 0\}.$$

A nice corollary of Theorem 3.1 follows.

Corollary 3.2. *Since the unknot is the only knot with genus 0, if a knot $K \subset S^3$ has the same $\widehat{\mathrm{HFK}}$ as the unknot U, then $K = U$* [OS04a].

3.3. **Knot Floer homology concordance invariants.** In this section we will define τ and ϵ in terms of Heegaard Floer homology: τ is a homomorphism from the rational homology concordance group $\mathcal{C}_\mathbb{Q}$ to $\mathbb{Z}$ [OS04b, Ras03] and ϵ is a map from $\mathcal{C}_\mathbb{Q}$ to $\{-1, 0, 1\}$ [Hom14, Lev21].

3.3.1. *τ invariant.* In order to define τ, the information in $\widehat{\mathrm{HFK}}$ is insufficient. Thankfully, the $\mathbb{F}[U]$ module HFK^- variant of knot Floer homology contains all the necessary information [OST08]. We can show [Hom20] that $\mathrm{HFK}^-(K) \otimes_{\mathbb{F}[U]} \mathbb{F}[U, U^{-1}] = \widehat{\mathrm{HF}}(S^3) \otimes_{\mathbb{F}[U]} \mathbb{F}[U, U^{-1}] = \mathbb{F}[U, U^{-1}]$; by the structure theorem on finitely-generated modules over a PID, this means that $\mathrm{HFK}^-(K)$ has exactly one free summand. Suppose η is the element of highest Alexander grading that generates this summand. Then we denote $\tau(K) = -A(\eta)$. In other words [Hom17], for a knot $K \subset S^3$,

$$\tau(K) := -\max\{s \mid \exists x \in \mathrm{HFK}^-_*(K, s),\ U^j x \neq 0 \text{ for all } j \geqslant 0\}.$$

We can also define τ directly from $\widehat{CFK}(K)$, by defining

$$\tau(K) = \min\{s \in \mathbb{Z} | \iota : \widehat{CFK}(K, j \leqslant s) \to \widehat{CFK}(K) \text{ induces a surjection in homology } \}$$

We record the following two properties. First, τ is a homomorphism from the smooth concordance group to $\mathbb{Z}$ [OS04b, Ras03]. In particular it obstructs concordance: if $\tau(K_1) \neq \tau(K_2)$ then K_1 and K_2 are not concordant, and $\tau(K) = 0$ whenever K is slice. Second, recall that the

slice genus $g_4(K)$ is the minimum genus of a smoothly embedded orientable surface in D^4 with boundary K. Then

$$|\tau(K)| \leqslant g_4(K) \leqslant u(K),$$

where $u(K)$ is the unknotting number [OS04b, Ras03]. The second inequality follows from the standard construction of a genus $u(K)$ surface from a sequence of crossing changes unknotting K.

3.3.2. ϵ *invariant.* Defining ϵ requires us to recall the graphical encoding of CFK^∞. We assume throughout that the complex is reduced. In order to present ϵ we define *vertically simplified* and *horizontally simplified bases* of CFK^∞. Recall that we can break the differential of CFK^∞ into the horizontal differential ∂^{horz}, the part of the differential that preserves the Alexander filtration, and the vertical differential ∂^{vert}, the part of the differential that preserves the algebraic filtration. We present ϵ as Hom does in [Hom17].

Definition 3.3. Let $\{x_i\}$ be a filtered basis for a reduced complex $CFK^\infty(K)$. The basis is *vertically simplified* if for each x_i exactly one of the following holds:

- $\partial^{\text{vert}} x_i = x_j$ for a unique x_j, and x_i is not in the image of ∂^{vert};
- $x_i = \partial^{\text{vert}} x_j$ for a unique x_j, and $\partial^{\text{vert}} x_i = 0$;
- x_i lies in the kernel but not the image of ∂^{vert}.

Proposition 3.4 ([Lev21, Proposition 2.4])**.** *There exists a pair of bases* $\{\xi_0, \dots, \xi_{2n}\}$ *and* $\{\eta_0, \dots, \eta_{2n}\}$ *of* $CFK^-(K)$ *such that* $\{\xi_0, \dots, \xi_{2n}\}$ *is a vertically simplified basis with a vertical arrow from* $\xi_{2j-1} \to \xi_{2j}$*, and* $\{\eta_0, \dots, \eta_{2n}\}$ *is a horizontally simplified basis with a horizontal arrow from* $\eta_{2j-1} \to \eta_{2j}$*. Moreover, these two bases are related by a change of coordinates such that* ξ_i *is a linear combination of* $\sum_j a_{ij}\eta_j$ *such that the only nonzero* a_{ij} *occur when* ξ_i *and* η_j *are at the same Alexander filtration level. The same holds for the inverse change of coordinates.*

Definition 3.5. We call ξ_0 the *vertically distinguished element* and η_0 the *horizontally distinguished element*. These are characterized by having no vertical (resp. horizontal) arrows coming into or out of them in the vertical (resp. horizontal) complex.

Remark 3.6. Note that from this description, it is easy to calculate the homology of either complex: it is $\mathbb{F}[U]$, generated by ξ_0 or η_0. Since the horizontal arrows are precisely the arrows left in $gCFK^-(K)$, the $\mathbb{F}[U]$ summand generated by η_0 is the same as the $\mathbb{F}[U]$ summand in $\text{HFK}^-(K)$; so $\tau(K) = -A(\eta_0)$. Similarly, the vertical complex is precisely $\widehat{CFK}$ since it is obtained by setting $U = 0$, thus

$$\tau(K) = A(\xi_0) = \min\{s \in \mathbb{Z} | \iota : \widehat{CFK}(K, j \leqslant s) \to \widehat{CFK}(K) \text{ induces a surjection in homology}\}$$

Hence $\tau(K) = A(\xi_0) = -A(\eta_0)$.

In [Hom14, Lemma 3.2], Hom proves that the horizontally distinguished element η_0 is an element ξ_i of some vertically simplified basis, and vice-versa. Now, we have three possibilities: either $\eta_0 = \xi_0$, so it has no vertical arrows, or $\eta_0 = \xi_1$ (with reordering), so it has an outgoing vertical arrow, or $\eta_0 = \xi_2$, so it has an incoming vertical arrow.

Definition 3.7. Let $\epsilon(K) \in \{-1, 0, 1\}$ according to the following rule. If η_0 has an incoming vertical arrow, then $\epsilon(K) = 1$, if η_0 has an outgoing vertical arrow, then $\epsilon(K) = -1$, otherwise if $\eta_0 = \xi_0, \epsilon(K) = 0$.

It remains open whether or not there always exists a simultaneously vertically and horizontally simplified basis of $CFK^\infty(K)$ for any knot $K \subset S^3$; however, thanks to [Hom14, Lemmas 3.2 and 3.3], there exists a horizontally simplified basis $\{\eta_i\}$ for $CFK^\infty(K)$ such that some η_j is ξ_0. We define ϵ using this choice of horizontally simplified basis.

Definition 3.8. For any knot $K \subset S^3$, let $\{\eta_i\}$ be a horizontally horizontally simplified basis for $CFK^\infty(K)$ such that some η_j is ξ_0. Then ϵ is defined as follows:

- $\epsilon(K) = -1$ if η_j is not in the kernel of ∂^{horz}.
- $\epsilon(K) = 0$ if η_j is in the kernel but not the image of ∂^{horz}.
- $\epsilon(K) = 1$ if η_j is in the image of ∂^{horz}.

Graphically, this implies that $\epsilon(K) = -1$ if η_j has an outgoing arrow, $\epsilon(K) = 1$ if η_j has an incoming arrow, and $\epsilon(K) = 0$ if η_j has no incoming or outgoing horizontal or vertical arrows.

Proposition 3.9 ([Hom14], Proposition 3.6)**.** *The invariant ϵ satisfies the following properties. Let $K, K_1, K_2 \subset S^3$ be knots, then:*

(1) If K is slice, then $\epsilon(K) = 0$.
(2) If $\epsilon(K) = 0$, then $\tau(K) = 0$.
(3) $\epsilon(-K) = -\epsilon(K)$.
(4) If $\epsilon(K_1) = \epsilon(K_2)$, then $\epsilon(K_1 \# K_2) = \epsilon(K_1)$. If $\epsilon(K_1) = 0$, then $\epsilon(K_1 \# K_2) = \epsilon(K_2)$.

4. Bordered Heegaard Floer Homology

Defined by Lipshitz, Ozsváth, and Thurston [LOT18], bordered Heegaard Floer homology is an enhancement of Heegaard Floer homology associating homology groups to any compact oriented three-manifold Y with boundary. We follow Hom's presentation of bordered Heegaard Floer homology as in [Hom14].

Let Y be a closed 3-manifold and let F be an abstract closed surface together with a null homologous embedding in Y. We can then decompose Y along F into two pieces Y_1 and Y_2 such that $\partial Y_1 \cong -\partial Y_2 \cong F$. This gives an orientation preserving diffeomorphism from F to ∂Y_1, and an orientation reversing diffeomorphism from F to ∂Y_2. A 3-manifold with a diffeomorphism (up to isotopy) from a standard surface to its boundary is called a bordered 3-manifold. To the closed surface F, we associate a differential graded algebra $\mathcal{A}(F)$. To Y_1, we associate the invariant $\widehat{CFA}(Y_1)$, which will be a right $\mathcal{A}_\infty$-module over the algebra $\mathcal{A}(F)$, while to Y_2 we associate the invariant $\widehat{CFD}(Y_2)$, which will be a Type D structure (see Definition 4.2). To a knot K_1 in Y_1, we may associate either $\widehat{CFA}(Y_1, K_1)$, a filtered $\mathcal{A}_\infty$-module, or $CFA^-(Y_1, K_1)$, an $\mathcal{A}_\infty$-module over $\mathcal{A}(F)$ with ground ring $\mathbb{F}[U]$, where U is a formal variable. For a satellite knot, $Y_1 = S^1 \times D^2$, $P \subset Y_1$ is the knot, and $Y_2 = S^3 \backslash \nu(K)$, the complement of the companion. Using [LOT18, Theorem 11.19], we can combine the bordered Heegaard Floer invariants of (Y_1, K_1) and Y_2 (glued along their boundary torus) to obtain $gCFK^-(P(K))$ and compute τ of the satellite.

4.1. Algebraic preliminaries.

We present the algebraic fundamentals underlying bordered Heegaard Floer homology, such as $\mathcal{A}_\infty$ algebras, $\mathcal{A}_\infty$-modules, the Type D structures of [LOT18], and the box tensor product. For a more detailed description, we refer the reader to [LOT18].

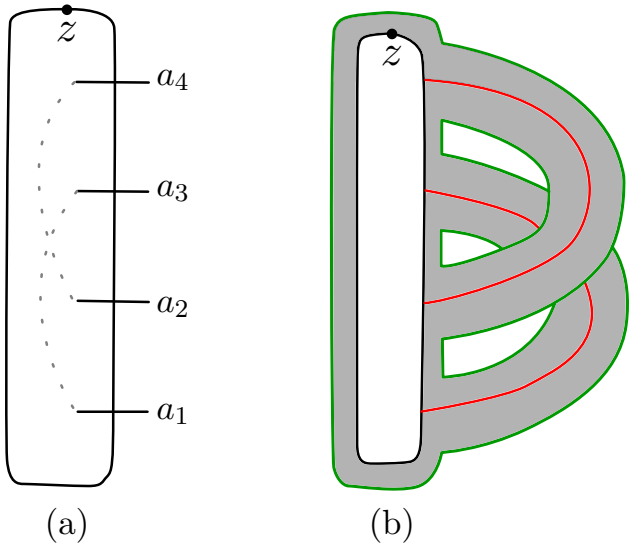


FIGURE 5. Sub-Figure (a) contains the pointed matched circle for the torus and sub-Figure (b) is the pointed matched surface for the torus. The surface is obtained by thickening the pointed matched circle and attaching 1-handles corresponding to the matchings $\{a_1, a_3\}$ and $\{a_2, a_4\}$. Notice that attaching a disc along the green boundary of the surface produces a punctured torus.

Definition 4.1. An A_∞ *algebra* over $\mathbb{F}$ is an $\mathbb{F}$-vector space $\mathcal{A}$ with a family of maps $\mu_i \colon \mathcal{A}^{\otimes i} \to \mathcal{A}$, $i \geqslant 1$, satisfying:

$$\sum_{i+j=n+1} \sum_{k=0}^{n-j} \mu_i\big(\mathbb{I}^{\otimes k} \otimes \mu_j \otimes \mathbb{I}^{\otimes n-j-k}\big) = 0,$$

for all $n \geqslant 1$. A differential graded algebra is the special case where $\mu_i = 0$ for $i \geqslant 3$, with μ_1 the differential and μ_2 the multiplication; an ordinary associative algebra is the further special case $\mu_1 = 0$. In this paper, we will only be concerned with the *torus algebra* $\mathcal{A}(T^2)$. An $\mathcal{A}_\infty$-algebra $\mathcal{A}$ is *strictly unital* if there is an element $1 \in \mathcal{A}$ with the property that $\mu_2(a, 1) = \mu_2(1, a) = a$ and $\mu_i(a_1, \ldots, a_i) = 0$ if $i \neq 2$ and $a_j = 1$ for some j.

In the torus case $F = T^2$, the torus algebra $\mathcal{A}(T^2)$ is just an associative graded algebra since it has vanishing differential. In order to specify the gluing of the boundaries of Y_1 and Y_2, we identify a meridian and a longitude of the torus. In the torus case, we specify a handle decomposition by a disc with two 1-handles attached such that the resulting boundary is connected and can be capped off with a disc. For technical reasons, we also place a z basepoint somewhere along the boundary of the disc.

More specifically, we represent this by a *pointed matched circle* $(\mathcal{Z}, z, \{a_1, a_3\}, \{a_2, a_4\})$, representing the boundary of the disc with labeled points denoting where to attach the attaching spheres of the 1-handles. Thus, the pointed matched circle $\mathcal{Z}$ consists of an oriented circle with five disjoint

marked points: a_1, a_2, a_3, a_4, and z, in that order as we traverse the circle in the clockwise direction. For the torus, in order to ensure that the boundary of the disc with the two 1-handles attached is connected, the points a_1 and a_3 are the endpoints of the handle whose core we denote as the arc α_1, and the points a_2 and a_4 are the endpoints of the handle whose core we denote as the arc α_2.

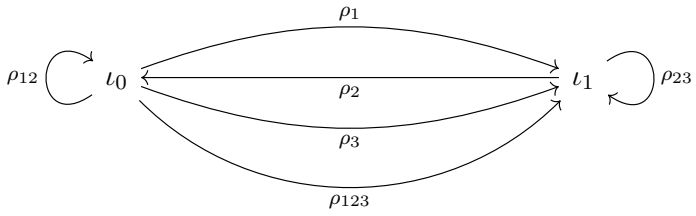


FIGURE 6. The torus algebra $\mathcal{A}(T)$ as a quiver.

Specifically, to the torus we associate the graded algebra $\mathcal{A}(T)$ with two minimal idempotents ι_0 and ι_1 forming the *ring of idempotents* $\mathbb{F}\iota_0 \oplus \mathbb{F}\iota_1 \subset \mathcal{A}(T)$, and 6 generators $\rho_1, \rho_2, \rho_3, \rho_{12}, \rho_{23}$, and ρ_{123} with vanishing differential and the following non-zero products:

$$\rho_1\rho_2 = \rho_{12}, \quad \rho_2\rho_3 = \rho_{23}, \quad \rho_1\rho_{23} = \rho_{123}, \quad \rho_{12}\rho_3 = \rho_{123}.$$

In particular, ρ_1, ρ_2, and ρ_3 are called Reeb chords and they are the following arcs on the pointed matched circle. In Figure 5, ρ_1 refers to the arc of $\mathcal{Z} - \{z\}$ between a_1 and a_2, ρ_2 the arc between a_2 and a_3, and ρ_3 the arc between a_3 and a_4. Note that there is no Reeb chord for the arc between a_4 and a_1 since the z-basepoint lies on this arc. There are compatibility relations between the idempotents and generators obtained by following paths in the pointed matched circle:

$$\rho_1 = \iota_0\rho_1\iota_1, \qquad \rho_2 = \iota_1\rho_2\iota_0, \qquad \rho_3 = \iota_0\rho_3\iota_1,$$

$$\rho_{12} = \iota_0\rho_{12}\iota_0, \qquad \rho_{23} = \iota_1\rho_{23}\iota_1, \qquad \rho_{123} = \iota_0\rho_{123}\iota_1.$$

Definition 4.2. A *type D structure over* $\mathcal{A}$ is an $\mathbb{F}$-vector space N, equipped with a left $\mathcal{I}$-action satisfying

$$N = \iota_0 N \oplus \iota_1 N,$$

and a map

$$\delta_1 : N \to \mathcal{A} \otimes_{\mathcal{I}} N,$$

satisfying the type D-relations

$$(\mu \otimes \mathrm{id}_N) \circ (\mathrm{id}_{\mathcal{A}} \otimes \delta_1) \circ \delta_1 = 0,$$

where $\mu : \mathcal{A} \otimes_{\mathcal{I}} \mathcal{A} \to \mathcal{A}$ is the multiplication map on $\mathcal{A}$. This inductively defines the maps

$$\delta_k : N \to \mathcal{A}^{\otimes k} \otimes_{\mathcal{I}} N,$$

by setting

$$\delta_0 = \mathrm{id}_N, \qquad \delta_i = \mathrm{id}_{\mathcal{I}}^{\otimes i-1} \otimes \delta_1 \circ \delta_{i-1}.$$

We say that N is *bounded* if there exists an integer n such that $\delta_i = 0$ for all $i > n$. Now, let us write $M[n]$ to denote a grading shift by n, i.e. $M[n]_d = M_{d-n}$.

Definition 4.3. A *(right unital) $\mathcal{A}_\infty$-module* is an $\mathbb{F}$-vector space M equipped with a right $\mathcal{I}$-action such that

$$M = M\iota_0 \oplus M\iota_1,$$

and a family of maps

$$m_i : M \otimes \mathcal{A}^{\otimes i-1} \to M[2-i], \quad i \geqslant 1$$

that satisfies the $\mathcal{A}_\infty$-relations

$$\begin{aligned} 0 = &\sum_{i=0}^{n} m_{n-i+1}\Big(m_{i+1}(x \otimes a_1 \otimes \cdots \otimes a_i) \otimes a_{i+1} \otimes \cdots \otimes a_n\Big) \\ &+ \sum_{i=1}^{n-1} m_n\Big(x \otimes a_1 \otimes \cdots \otimes a_{i-1} \otimes a_i a_{i+1} \otimes a_{i+2} \otimes \cdots \otimes a_n\Big). \end{aligned}$$

and the unital conditions

$$m_2(x,1) = x, \qquad m_i(x,\ldots,1,\ldots) = 0, \quad i > 2.$$

We say that M is *bounded* if there exists an integer n such that $m_i = 0$ for all $i > n$.

Given a type D structure N over $\mathcal{A}$ and an $\mathcal{A}_\infty$-module M, we may define the *box tensor product* $M \boxtimes N$ to be the $\mathbb{F}$-vector space $M \otimes_{\mathcal{I}} N$ equipped with the differential

$$\partial^{\boxtimes}(x \otimes y) = \sum_{k=0}^{\infty} (m_{k+1} \otimes \mathrm{id}_N)(x \otimes \delta_k(y)).$$

The sum is finite if at least one of M or N is bounded.

4.2. **Bordered Heegaard Floer homology and pairing.** Consider two compact, oriented 3-manifolds Y_1, Y_2 with torus boundary. Let $K_1 \subset Y_1$ be a knot. We may associate two different algebraic invariants to (Y_1, K_1) and Y_2: the $\mathcal{A}_\infty$-module $CFA^-(Y_1, K_1)$, and the type D structure $CFD(Y_2)$. The former is an $\mathcal{A}_\infty$-module over the torus algebra $\mathcal{A}(T^2)$ with ground ring $\mathbb{F}[U]$, where U is a formal variable introduced to keep track of the multiplicity of holomorphic discs at the w-basepoint in the doubly-pointed Heegaard diagram of K_1.

In [LOT18], Lipshitz, Ozsváth, and Thurston prove that the box tensor product of the $\mathcal{A}_\infty$-module associated to a pattern knot and the type D structure associated to a companion knot recover the associated graded complex of their satellite. This is known as the *pairing theorem* in the literature.

Theorem 4.4 ([LOT18, Theorem 11.19])**.** *Let Y_1 and Y_2 be compact, oriented 3-manifolds with parametrized boundary, and let*

$$Y = Y_1 \cup_f Y_2$$

be obtained by gluing their boundaries by an orientation-reversing diffeomorphism. Let $K_1 \subset Y_1$ be a knot with null-homologous image $K \subset Y$ after the gluing. Then there is a homotopy equivalence of $\mathbb{F}[U]$-complexes

$$gCFK^-(Y,K) \simeq CFA^-(Y_1,K_1) \boxtimes \widehat{CFD}(Y_2).$$

In particular, in the case where $Y_1 = S^1 \times D^2$, $K_1 = P$ is a pattern knot in the solid torus, and $Y_2 = S^3 \backslash \nu(K)$ where $K \subset S^3$ is a companion knot, we have

$$gCFK^-(S^3, P(K)) \simeq CFA^-(S^1 \times D^2, P) \boxtimes \widehat{CFD}(S^3 \backslash \nu(K)).$$

4.3. **Constructing $CFA^-(S^1 \times D^2, P)$.** In this section, we will describe how one can obtain $CFA^-(S^1 \times D^2, P)$ from a doubly pointed bordered Heegaard diagram $\mathcal{H}$ for $P \subset S^1 \times D^2$.

Definition 4.5. For a three-manifold Y with $\partial Y = T^2$, a *doubly pointed bordered Heegaard diagram* for Y is a tuple $\mathcal{H} = (\overline{\Sigma}, \alpha^c, \alpha^a, \beta, z, w)$ such that:

- $\overline{\Sigma}$ is a compact oriented surface of genus g with a single boundary component,
- there is a $(g-1)$-tuple of pairwise disjoint circles $\alpha^c = (\alpha_1^c, \dots, \alpha_{g-1}^c)$ in ${\overline{\Sigma}}^o$ denoting the interior of $\overline{\Sigma}$,
- there is a pair of disjoint arcs $\alpha^a = (\alpha_1^a, \alpha_2^a)$ in $\overline{\Sigma} \backslash \alpha^c$ with endpoints on $\partial\overline{\Sigma}$,
- there is a g-tuple of pairwise disjoint circles $\beta = (\beta_1, \dots, \beta_g)$ in ${\overline{\Sigma}}^o$,
- z and w are basepoints in $\overline{\Sigma} \backslash (\alpha^a \cup \alpha^c \cup \beta)$, with $z \in \partial\overline{\Sigma}$.

In addition, we require that the α-curves and β-curves intersect transversely, that the complement of the α-curves and the complement of the β-curves be connected, and that $(\partial\overline{\Sigma}, z, \partial\alpha_1^a, \partial\alpha_2^a)$ describes a pointed matched circle as specified above.

Whenever the context is clear, we refer to doubly pointed bordered Heegaard diagrams simply as bordered Heegaard diagrams. Similarly to standard knot Floer homology, a bordered Heegaard diagram specifies a knot embedded in Y by connecting w to z in the complement of the β-curves and connecting z to w in the complement of the α-arcs. Here if the two sets of arcs cross, the convention is that the second set crosses under the first. Every knot in a bordered three-manifold can be realized by a doubly pointed Heegaard diagram.

A pattern knot embedded in the solid torus $V = S^1 \times D^2$ is a $(1,1)$-pattern knot if it admits a bordered Heegaard diagram of genus one [Che23]. We will focus exclusively on $(1,1)$ patterns in this paper. The arc α_1^a represents a meridian $\mu_V = \{\text{pt}\} \times \partial D^2$, and the arc α_2^a represents a longitude $\lambda_V = S^1 \times \{\text{pt}\}$.

Genus one diagrams have two α arcs, no α circles, and one β curve. Denote by $\mathfrak{S}(\mathcal{H})$ the set of intersection points of the α-arcs and the β-curve. As an $\mathbb{F}[U]$-vector space, $CFA^-(S^1 \times D^2, P)$ is generated by elements of $\mathfrak{S}(\mathcal{H})$, equipped with the right $\mathcal{I}$-action given by

$$x \cdot \iota_0 = \begin{cases} x & \text{if x occupies the arc } \alpha_1^a \\ 0 & \text{otherwise} \end{cases}$$

$$x \cdot \iota_1 = \begin{cases} x & \text{if x occupies the arc } \alpha_2^a \\ 0 & \text{otherwise.} \end{cases}$$

The $\mathcal{A}_\infty$-structure is given by maps (suppressing grading shifts)

$$m_{j+1} : CFA^-(S^1 \times D^2, P) \otimes \mathcal{A}^{\otimes j} \to CFA^-(S^1 \times D^2, P),$$

which are defined by

$$m_{j+1}(x, \rho_{i_1}, \dots, \rho_{i_j}) = U^{n_w(\phi)} y, \tag{5}$$

where ϕ denotes a particular pseudoholomorphic disc.

To compute this, we draw the bordered Heegaard diagram on the square fundamental domain of this torus. The corners of this square are cut out and replaced by the boundary circle. By convention, the z basepoint is placed on the top-left corner. Then, we pass to the universal cover of the Heegaard diagram $\mathcal{H}$ and count pseudoholomorphic discs ϕ which are bounded by an α-arc and β-curve that intersect at non-reflex angles (the condition on angles gives the right Maslov index). We then label the two intersection points by x and y such that the α-arc is immediately encountered if we travel counterclockwise from x along the boundary. Identifying $(\partial\overline{\Sigma}, z, \partial\alpha_1^a, \partial\alpha_2^a)$ with our pointed matched circle $\mathcal{Z}$, we denote any parts of $\mathcal{Z}$ that are present in the boundary of ϕ by their corresponding generators $\rho_{i_1}, \dots, \rho_{i_j}$, where we apply the product relations where possible. The number $n_w(\phi)$ denotes the number of copies of the w-basepoint contained in ϕ. We also insist that ϕ not intersect the top left corner of $\mathcal{H}$, to ensure that ϕ does not contain the z-basepoint.

4.4. **Constructing** $CFD(S^3\backslash\nu(K))$**.** Let $K \subset S^3$ be a fixed companion knot. Given a basis of $CFK^-(K)$, [Lev21] describes a method of algorithmically computing the type D structure $CFD(S^3\backslash\nu(K))$, first described in [LOT18, Theorems 11.26 and A.11].

Theorem 4.6 ([Lev21, Theorem 2.6])**.** *Let $K \subset S^3$ be a knot. Given bases $\{\widetilde{\xi}_0, \dots, \widetilde{\xi}_{2n}\}$ and $\{\widetilde{\eta}_0, \dots, \widetilde{\eta}_{2n}\}$ satisfying the conclusions of Proposition 3.4, the type-D structure $\widehat{\mathrm{CFD}}(S^3\backslash\nu(K))$ satisfies the following properties:*

- *The summand $\iota_0 \cdot \widehat{\mathrm{CFD}}(S^3\backslash\nu(K))$ has dimension $2n+1$, with designated bases $\{\xi_0, \dots, \xi_{2n}\}$ and $\{\eta_0, \dots, \eta_{2n}\}$ related by*

$$\xi_p = \sum_{q=0}^{2n} a_{p,q}\eta_q \qquad \text{and} \qquad \eta_p = \sum_{q=0}^{2n} b_{p,q}\xi_q.$$

  These elements are all homogeneous with respect to the grading by relative spin^c *structures.*
- *The summand $\iota_1 \cdot \widehat{\mathrm{CFD}}(S^3\backslash\nu(K))$ has dimension*

$$\sum_{j=1}^{n}(k_j + l_j) + s, \qquad \text{where} \qquad s = 2|\tau(K)|,$$

  with basis

$$\bigcup_{j=1}^{n}\{\kappa_1^j, \dots, \kappa_{k_j}^j\} \ \cup \ \bigcup_{j=1}^{n}\{\lambda_1^j, \dots, \lambda_{l_j}^j\} \ \cup \ \{\mu_1, \dots, \mu_s\}.$$

- *For $j = 1, \dots, n$, corresponding to the vertical arrow $\widetilde{\xi}_{2j-1} \longrightarrow \widetilde{\xi}_{2j}$, there are coefficient maps*

$$\xi_{2j} \xrightarrow{D_{123}} \kappa_1^j \xrightarrow{D_{23}} \cdots \xrightarrow{D_{23}} \kappa_{k_j}^j \xleftarrow{D_1} \xi_{2j-1}. \tag{6}$$

- *For $j = 1, \dots, n$, corresponding to the horizontal arrow $\widetilde{\eta}_{2j-1} \longrightarrow \widetilde{\eta}_{2j}$, there are coefficient maps*

$$\eta_{2j-1} \xrightarrow{D_3} \lambda_1^j \xrightarrow{D_{23}} \cdots \xrightarrow{D_{23}} \lambda_{l_j}^j \xrightarrow{D_2} \eta_{2j}. \tag{7}$$

- *Depending on $\tau(K)$, there are additional coefficient maps*

$$\begin{cases} \eta_0 \xrightarrow{D_3} \mu_1 \xrightarrow{D_{23}} \cdots \xrightarrow{D_{23}} \mu_s \xleftarrow{D_1} \xi_0, & \tau(K) > 0, \\ \xi_0 \xrightarrow{D_{12}} \eta_0, & \tau(K) = 0, \\ \xi_0 \xrightarrow{D_{123}} \mu_1 \xrightarrow{D_{23}} \cdots \xrightarrow{D_{23}} \mu_s \xrightarrow{D_2} \eta_0, & \tau(K) < 0. \end{cases} \tag{8}$$

We call the subspaces of $CFD(S^3\backslash\nu(K))$ spanned by the generators specified in 6, 7, and 8 the *vertical chains*, *horizontal chains*, and *unstable chain*, respectively.

In general, we will be working with arbitrary companion knots K, and hence obtaining different unstable chains in the *CFD* structures based on the sign of $\tau(K)$ and $\epsilon(K)$. Since we only care about the summand containing the free $\mathbb{F}[U]$-summand generator in $CFA^-(S^1 \times D^2, P) \boxtimes CFD(S^3\backslash\nu(K))$ when computing $\tau(P(K))$ and $\epsilon(P(K))$, it turns out that it suffices to consider a neighborhood of the unstable chain rather than the entire type D structure itself, which we call the *unstable neighborhood*.

5. Heegaard diagrams for generalized Mazur patterns

To compute $CFA^-(Q_{i,(m,n)})$, we need its bordered Heegaard diagram. To obtain this, we use results from [Che23], using the correspondence between $(1,1)$ unknot patterns and two-bridge links, and then converting the two-bridge link into Chen's standard form using the Conway normal form.

5.1. **$(1,1)$-patterns and two-bridge links.** Recall that a $(1,1)$ unknot pattern is a knot $P \subset S^1 \times D^2$ admitting a genus 1 bordered Heegaard diagram such that $P(U) = U$. A link in S^3 is a *two-bridge link* if it has a diagram with exactly two local maxima. Every two-bridge link has one or two components, and each component is unknotted.

Proposition 5.1 ([Che23, Theorem 5.4])**.** *There is a bijective correspondence between two-component two-bridge links in S^3 and $(1,1)$ unknot patterns in $S^1 \times D^2$*

The idea behind this bijection is to consider one of the components of our two-component two-bridge link L as the meridian of the solid torus. Let the two components be L_1 and L_2. Since L_1 is unknotted, $S^3\backslash\nu(L_1)$ is a solid torus, so $L_2 \subset S^3\backslash\nu(L_1)$ is a pattern knot. Applying this pattern to the unknot is equivalent to gluing $\nu(L_1)$ back in to get back S^3. Since L_2 was an unknot in S^3, we get an unknot. Hence every two-component two-bridge link gives rise to an unknot pattern. To show that it is a $(1,1)$-pattern and that all $(1,1)$-patterns are exhausted this way requires some more work, done in [Che23, Theorem 5.4]. This work boils down to using the invariants we define below to prove injectivity and surjectivity.

5.2. **Schubert normal form and Conway normal form.** Two bridge links are described by a standard form called the *Schubert Normal Form* [Sch56].

Definition 5.2 ([Che23, §5.2])**.** The Schubert normal form of a two-bridge link is given by a pair of coprime integers p, q and written $b(p,q)$. These integers satisfy $p > 0$ and $0 \leqslant |q| \leqslant \frac{p}{2}$.

Proposition 5.3 ([Che23, §5.2])**.** *The Schubert normal form satisfies the following properties:*

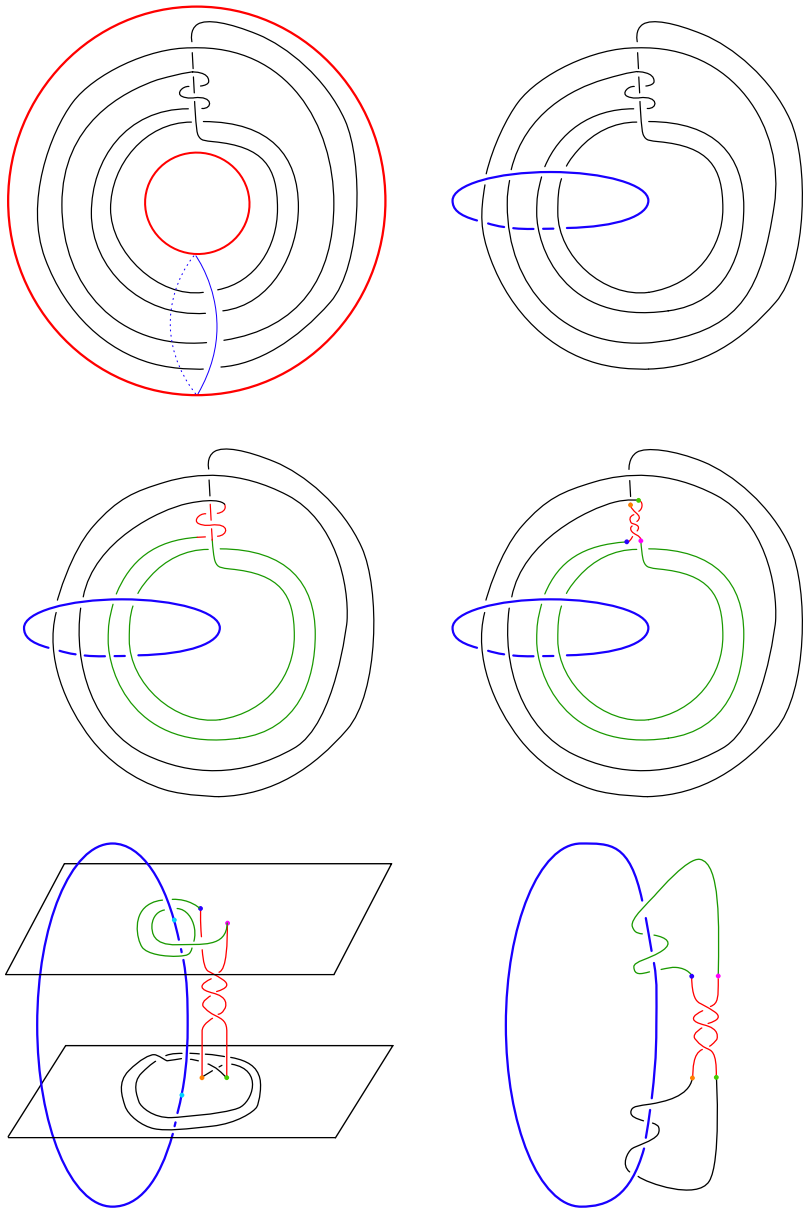


FIGURE 7. We show the isotopy between the two-bridge link form of $Q_{2,(2,2)}$ and note that this method generalizes to $Q_{i,(m,n)}$. Start with standard presentation of the pattern $Q_{2,(2,2)}$ in the solid torus. Consider the link formed by $Q_{2,(2,2)}$ and the blue torus meridian circle. Moving to three dimensions, bring the 2 green inner twists of $Q_{2,(2,2)}$ up and lower the two outer black twists. Pull the green and black arcs up and down to produce the two-bridge link $C(4,3,4)$. Notice that this procedure implies $Q_{i,(m,n)}$ is isotopic to $Q_{i,(n,m)}$.

(1) $b(p,q)$ *is a two-component link if and only if* p *is even*
(2) $b(p,q)$ *is the mirror image of* $b(p,-q)$
(3) $b(p,q)$ *and* $b(p',q')$ *are isotopic to each other (as unoriented links) if and only if* $p=p'$ *and* $q' \equiv q^{\pm 1} \pmod{p}$

While the Schubert normal form gives us an easy check to see if two patterns are the same, it is difficult to get to it from a picture. For that we use Conway's normal form [Con70].

Definition 5.4 (Conway normal form)**.** Let $a_1, \ldots, a_n$ be nonzero integers. Take four horizontal strands, numbered 1 to 4 from top to bottom. Reading from left to right, insert n twist regions. The i-th region has $|a_i|$ crossings. It lies between strands 2 and 3 if i is odd, and between strands 3 and 4 if i is even. Strand 1 has no crossings. The crossings in a region are all of the same type: if i is odd and $a_i > 0$, the strand running down to the right is on top; if i is even and $a_i > 0$, the strand running up to the right is on top. If $a_i < 0$, the crossings are reversed. Close the diagram on the left with two caps, joining strand 1 to strand 2 and strand 3 to strand 4. Close it on the right in the same way if n is odd. If n is even, join strand 2 to strand 3 and strand 1 to strand 4 on the right. The resulting link is denoted $C(a_1, \ldots, a_n)$, and the diagram is its Conway normal form.

It can be deduced from Figure 7 that the two-bridge link form of $Q_{i,(m,n)}$ is $C(2m, 2i-1, 2n)$ in Conway normal form. This two bridge form has $2m$ positive half-twists between the middle two strands, then $2i-1$ negative half-twists between the rightmost two strands, then $2n$ positive half-twists again between the middle two strands.

We then follow Chen's methodology and translate the Conway normal form of the two-bridge link to the Schubert normal form [Sch56]. One can convert any two-bridge link represented by $C(a_1, \ldots, a_n)$ to $b(p,q)$ by using the following iterated fraction formula [PX24]:

$$\frac{q}{p} = \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots_{+\frac{1}{a_n}}}}}$$

In order to convert the other way, for a given $b(p,q)$ we simply expand as a continued fraction. Thus we can expand $C(2m, 2i-1, 2n)$ to obtain:

$$\frac{q}{p} = \frac{1}{2n + \frac{1}{2i-1+\frac{1}{2m}}} = \frac{1+4im-2m}{2n+2m-4mn+8imn}.$$

Thus we have that the Schubert normal form for the pattern $Q_{i,(m,n)}$ is $b(p,q)$ with $p = 2n+2m-4mn+8imn$, $q = 1+4im-2m$.

5.3. **Notation for Heegaard diagrams.** Every $(1,1)$ pattern knot admits a genus-one doubly pointed bordered Heegaard diagram. Chen classifies diagrams of unknot patterns using a two integer parametrization (r,s) with $r \neq 0$ and $s \geqslant 0$ satisfying $\gcd(2|r|-1, s+1) = 1$ [Che23]. These Heegaard diagrams are denoted $H(r,s)$. Note that the gcd condition ensures that the β-curve is connected [Che23, Lemma 5.1].

We draw the unit square with opposite sides identified producing the torus. The vertical edges of the square are the meridian μ of the torus, and the horizontal edges are the longitude λ. The

diagram $H(r,s)$ is then determined by how we draw the β-curve and the two basepoints w and z. The vertical edges cut the β-curve into arcs of two kinds: a *rainbow* is an arc whose endpoints lie on the same side of the square and a *strand* is an arc whose endpoints lie on opposite sides of the square.

Following Chen's notation, we split the square into a *main region* and a *dependent region*, separated by the horizontal dashed line in Figure 8. The main region is drawn following a simple procedure for which drawing the dependent region requires no choices; however, it is specified by a non-trivial computation. The main region contains both basepoints, and the part of the β-curve in the main region is a single arc called the *main arc*. The part of the β-curve inside the dependent region is also a single arc called the *dependent arc*.

The β-curve must satisfy two conditions for $H(r,s)$ to be a diagram of an unknot pattern. Removing the z basepoint, what remains must be a Heegaard diagram for S^3 as seen in Figure 4. Thus we need β to be isotopic to μ in the complement of w. The first condition concerns the horizontal edges, in particular $H(r,s)$ is a diagram for S^3 if and only if $\beta \cdot \lambda = \pm 1$. For any other value it represents a lens space or $S^1 \times S^2$. As specified by Chen's construction of $H(r,s)$, the β-curve meets the horizontal edges in exactly one point. Conversely, Chen shows that the diagram of any $(1,1)$ unknot pattern can be isotoped, in the complement of w and z, so that β meets λ exactly once [Che23, Lemma 5.2]. The second condition concerns the vertical edges. Suppose $\beta \cdot \lambda = \pm 1$ but $\beta \cdot \mu = k \neq 0$. Then β is not isotopic to μ, and the curve that bounds a disk in the torus is no longer the meridian of the standard parametrization. Therefore, we require that $\beta \cdot \mu = 0$.

The dependent arc is there to ensure that $\beta \cdot \mu = 0$ and is determined by r and s. The horizontal edges and the dashed line cut the torus into two annuli. An embedded arc in an annulus with fixed endpoints is determined, up to isotopy, by its signed number of intersections with μ. Since $\beta \cdot \mu = 0$, the intersections of the dependent arc with μ must cancel the intersections of the main-region arc with μ. Chen gives the following algorithm for drawing the $H(r,s)$ diagram. We describe the case $r > 0$ and note that the case $r < 0$ is obtained by reflecting the diagram, as shown in Figure 8.

(1) Start with a square whose opposite sides are identified to produce a torus. The vertical sides are the α-arc μ and the horizontal sides are the α-arc λ (both drawn in red). Draw the horizontal dashed line. The main region lies below it and the dependent region lies above it.
(2) Mark $2r+s+1$ points on the left side, below the dashed line, and label them $0, 1, \ldots, 2r+s$ from bottom to top. Mark points at the same heights on the right side, with the same labels. Thus, points with the same label are identified in the torus. All β-arcs (drawn in blue) in the main region will end at these points.
(3) Draw r concentric rainbows on the left side, joining the left points i and $2r-1-i$ for $0 \leqslant i \leqslant r-1$. The outermost rainbow starts at the bottom left of the square. Place the z-basepoint inside the innermost rainbow.
(4) Draw s strands, joining the left point $2r-1+j$ to the right point j for $1 \leqslant j \leqslant s$. They start above the rainbows on the left and end a little above the bottom right of the square.
(5) Draw r concentric rainbows on the right side, joining the right points $s+i$ and $s+2r+1-i$ for $1 \leqslant i \leqslant r$. The outermost rainbow starts just above the strands. Place the w-basepoint

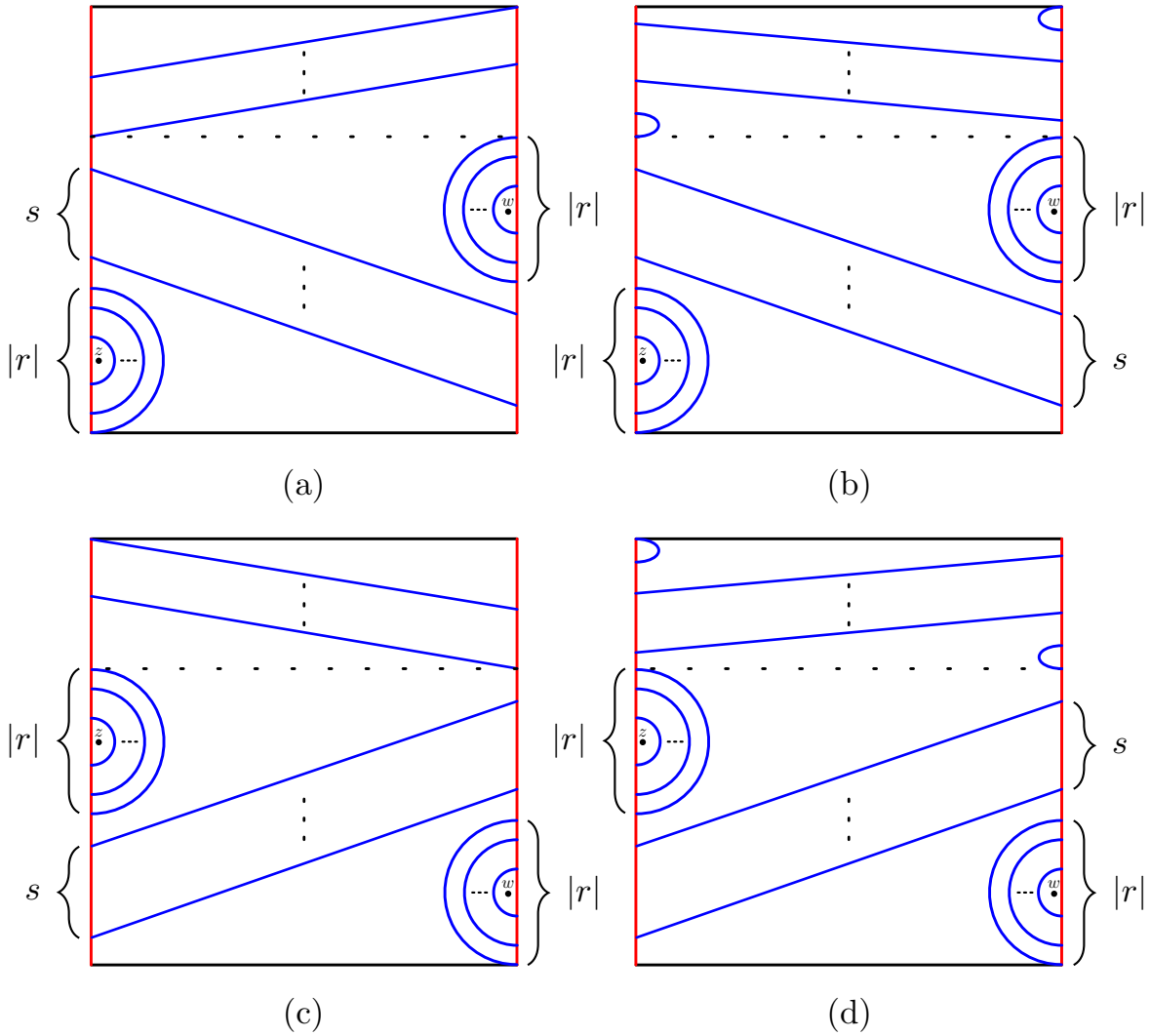


FIGURE 8. Diagrams (a) and (b) correspond to the (1,1) diagrams $H(r,s)$ with $r > 0$. Diagrams (c) and (d) correspond to the (1,1) diagrams $H(r,s)$ with $r < 0$. The horizontal dotted line across each diagram separates the diagram into the main region (bottom) and dependent region (top). It is easy to see that the only differences between (a) and (b) lie in the dependent region, and similarly for (c) and (d). These differences are there to ensure that the algebraic intersection number between the α-curve and β-curve is correct.

inside the innermost rainbow. This concludes the drawing of the main region. We denote the β-arc contained in this region by l_m.

(6) Two points are still unused: the left point $2r+s$, at the top of the main region, and the right point 0, at the bottom of the square. These will be the endpoints of the dependent arc.

(7) Draw the dependent arc. It starts at the left point $2r+s$ and enters the dependent region. There it winds around the torus horizontally $|\sigma(r,s)|$ times, with:

$$\sigma(r,s) \;=\; \sum_{j=1}^{2r+s-1} (-1)^{\left\lfloor \frac{j(2r-1)}{2r+s} \right\rfloor} \;-\; 1.$$

Each time it reaches λ it continues from the same height on the opposite side. Furthermore, the direction of winding is the one that makes the total signed number of intersections of β with μ in the whole diagram equal to zero. The arc then crosses the top left side of the square exactly once and continues from the bottom side, and ends at the right point 0. The result is a single closed curve by the condition $\gcd(2r-1, s+1) = 1$. We denote the β-arc contained in the dependent region by l_d.

Remark 5.5. We note that Chen's expression for σ in the statement of Proposition 6.2 is incorrect; however, the expression we use is correct and is derived during the proof of Proposition 6.2 [Che23]. The mistake lies in the proof's last equality. We hope this remark helps avoid confusion surrounding our result.

5.4. **Schubert normal form to Heegaard diagram.** Chen [Che23, Theorem 5.4] states that if a two bridge link $b(p,q)$ corresponds to the Heegaard diagram $H(r,s)$ then we must have:

$$p = 2s + 4|r| \qquad q = \operatorname{sign}(r)(2|r| - 1).$$

Solving these equations for:

$$p = 2n + 2m - 4mn + 8imn \quad \text{and} \quad q = 1 + 4im - 2m,$$

we get that:

$$r = 1 + 2im - m \quad \text{and} \quad s = n + 3m - 2mn + 4imn - 4im - 2$$

Therefore, in Chen's notation we have that the Heegaard diagram corresponding to $Q_{i,(m,n)}$ is $H(1 + 2im - m, n + 3m - 2mn + 4imn - 4im - 2)$.

5.5. **Determining the dependent region.** To draw the diagram for $Q_{i,(m,n)}$, we need to describe the dependent region, which will be as in (a) of Figure 8.

Proposition 5.6. *For $i, m, n \geqslant 1$, the Heegaard diagram for $Q_{i,(m,n)}$ has exactly $m+n-2$ strands in the dependent region i.e. $l_d \cdot \mu = m + n - 2$.*

Proof. Since the number of dependent strands needs to be equal to $-l_m \cdot \mu$, this is equivalent to the statement $l_m \cdot \mu = 2 - m - n$. By the corrected expression of Chen's Proposition 6.2 [Che23] (Remark 5.5) this is equivalent to showing that:

$$\sigma(r,s) + 1 = \sum_{j=1}^{2r+s-1} (-1)^{\left\lfloor \frac{j(2r-1)}{2r+s} \right\rfloor} = m + n - 1, \tag{9}$$

with $r = 1+2im-m$ and $s = n+3m-2mn+4imn-4im-2$. To prove Equation 9, we substitute $a = 2r+s = n+m-2mn+4imn$ and $b = 2r-1 = 4im-2m+1$ and obtain:

$$\sigma(r,s)+1 = \sum_{j=1}^{a-1}(-1)^{\lfloor \frac{jb}{a} \rfloor}.$$

Note that $a = m+bn$ implies $\gcd(a,b) = \gcd(m,b) = \gcd(m, 1-2m+4im) = 1$, so $\frac{jb}{a}$ will never be an integer. Furthermore, note that $\left\lfloor \frac{jb}{a} \right\rfloor$ will take on every value from 0 to $b-1$ as j goes from 1 to $a-1$. For a given integer k, let $N(k)$ be the number of indices j such that $\left\lfloor \frac{jb}{a} \right\rfloor = k$; notice that these are exactly the j that satisfy $k < \frac{jb}{a} < k+1$. These are strict inequalities since we showed that $\frac{jb}{a}$ can never be an integer. This expression can be rewritten as:

$$\frac{ak}{b} < j < \frac{ak}{b} + \frac{a}{b}.$$

Therefore, we need to find the number of integers in the open interval $(\frac{ak}{b}, \frac{ak}{b} + \frac{a}{b})$ for a given integer k. Since $\frac{a}{b} = \frac{m}{b} + n$, the length of this open interval is not an integer. An open interval of non-integer length x contains either $\lfloor x \rfloor$ or $\lceil x \rceil$ integers. Since $m < b$, this interval must then contain n or $n+1$ integers. Define ϵ such that $N(k) = n + \epsilon(k)$ where $\epsilon(k) = 0$ or 1. Since there are $a-1$ total indices:

$$\sum_{k=0}^{b-1} N(k) = \sum_{k=0}^{b-1}(n+\epsilon(k)) = a-1 \implies bn + \sum_{k=0}^{b-1}\epsilon(k) = bn+m-1 \implies \sum_{k=0}^{b-1}\epsilon(k) = m-1.$$

Thus there are exactly $(m-1)$ integers k for which $N(k) = n+1$. Now, grouping the original sum $\sigma(r,s)+1 = \sum_{j=1}^{a-1}(-1)^{\lfloor \frac{jb}{a} \rfloor}$ by the value of $\left\lfloor \frac{jb}{a} \right\rfloor$, we obtain:

$$\sigma(r,s)+1 = \sum_{k=0}^{b-1} N(k)(-1)^k = \sum_{k=0}^{b-1}(-1)^k n + \sum_{k=0}^{b-1}(-1)^k \epsilon(k).$$

Since $b = 1+4im-2m$ is odd, $\sum_{k=0}^{b-1}(-1)^k = 1 + \sum_{k=1}^{b-1}(-1)^k = 1$. Hence we have:

$$\sigma(r,s)+1 = n + \sum_{k=0}^{b-1}(-1)^k \epsilon(k).$$

We claim that $\epsilon(k) = 1$ implies that k is even. Recall that for any real number p we denote $\{p\} := p - \lfloor p \rfloor$ as the fractional part of p. Looking at an open interval $(p, p+x)$ where x is not an integer, if we shift it to $(p-\{p\}, p+x-\{p\})$, this contains $\lfloor x \rfloor$ integers; sliding it back up by $\{p\}$ may or may not introduce a new integer. This happens if and only if $\{p\} > \{p+x\}$. Thus, $\epsilon(k) = 1$ if and only if $\left\{\frac{ak}{b}\right\} > \left\{\frac{a(k+1)}{b}\right\}$. Using $\frac{a}{b} = n + \frac{m}{b}$, we get that this condition is equivalent to $\left\{\frac{mk}{b}\right\} > \left\{\frac{mk}{b} + \frac{m}{b}\right\}$, which happens only if adding $\frac{m}{b}$ crosses an integer. Thus, $\frac{m}{b} + \left\{\frac{mk}{b}\right\} > 1$. Choose $q \in \mathbb{Z}$ and $y \in \mathbb{R}$ such that $mk = qb + y$, so $\left\{\frac{mk}{b}\right\} = \frac{y}{b}$. Then $\frac{m}{b} + \left\{\frac{mk}{b}\right\} > 1 \iff m + y > b \iff y > b - m$. Therefore we have that:

$$\epsilon(k) = 1 \iff (mk \mod b) > b - m.$$

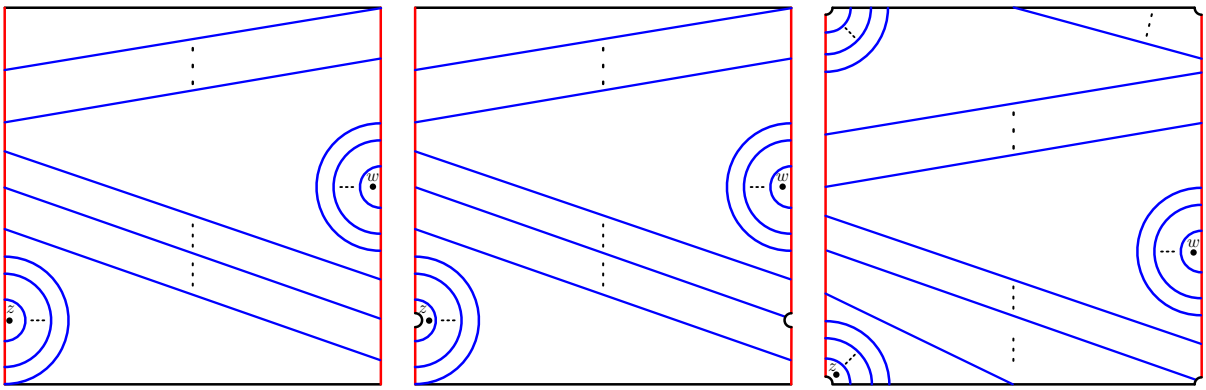


FIGURE 9. To go from the $H(r,s)$ Heegaard diagram for $Q_{i,(m,n)}$ displayed in (a) to a convenient bordered Heegaard diagram, puncture the torus near the z-basepoint as shown in (b) and then pull the β-curve down as shown in (c).

We know from earlier that this happens for exactly $m-1$ values of k. We explicitly find all $m-1$ distinct k that satisfy this condition. Let $k = (4i-2)q$ for $q = 1, \dots, m-1$. Then $mk = (4im-2m)q \equiv -q \mod b$ which is $b-q \mod b$. Since $q < m$, $b-q > b-m$. These are the exact $m-1$ values of k for which $\epsilon(k) = 1$, and $k = (4i-2)q$ is always even. Thus $\epsilon(k) = 1 \implies (-1)^k = 1$, so we obtain:

$$\sum_{k=0}^{b-1} (-1)^k \epsilon(k) = m-1 \implies \sigma(r,s) + 1 = n+m-1 \implies \sigma(r,s) = n+m-2.$$

Thus, the algebraic intersection number $-\sigma(r,s) = l_m \cdot \mu = -(m+n-2)$, so we must have $m+n-2$ dependent strands oriented positively to compensate as shown in subplot (a) of Figure 8. □

Therefore, we have shown that for any pattern knot $Q_{i,(m,n)}$, its corresponding Heegaard diagram will be $H(1+2im-m, n+3m-2mn+4imn-4im-2)$ in Chen's notation; from Proposition 5.6, its dependent region will contain $m+n-2$ strands with the orientation of (a) in Figure 8.

Finally, to convert from a standard Heegaard diagram to a bordered Heegaard diagram, we follow the steps described in Figure 9. The τ and ϵ computations in Theorem 1.2 are obtained using immersed curve techniques. We also use standard bordered Heegaard Floer homology techniques to recover the same τ computations and additionally compute $CFA^-(Q_{i,(2,2)})$ for $i \geqslant 1$, which may be of interest for future work on generalized Mazur patterns.

6. Immersed curves and computation

6.1. **Immersed curve techniques for computing bordered Heegaard Floer homology.**

Complete but still needs to be written up.

6.2. **Calculating invariants using immersed curve techniques.** τ, ε computation complete but write-up and Inkscape diagrams still not finished.

7. Extended computations for a sub-family of patterns via the pairing theorem

In this section, we compute $CFA^-(Q_{i,(2,2)})$ using the bordered Heegaard diagram for $Q_{i,(2,2)}$ obtained in section 5, the relevant portion of the associated graded complex $gCFK^-(Q_{i,(2,2)})$, and a closed formula for τ by following Levine's methodology [Lev21].

7.1. **Preliminaries.** Before we compute $CFA^-(Q_{i,(2,2)})$, we recall how to pair such a CFA^- structure with an arbitrary $\widehat{CFD}(K)$ in order to compute the relevant portion $gCFK^-(Q_{i,(2,2)}(K))$. To justify that our computation is constrained to the relevant part, we need the following preliminary results.

Lemma 7.1 ([Lev21, Lemma 2.7])**.** *In $\widehat{CFD}(S^3\backslash\nu(K))$ for any knot $K\subset S^3$, we have $(D_1\circ D_2\circ D_3)(\xi_0)=0$ (where ξ_0 is the vertically distinguished element).*

This lemma will ensure that $\rho_3\rho_2\rho_1$ elements in the CFA^- structure do not pair with any arrows in the $\widehat{CFD}$ structure, thus allowing us to restrict our computation to relevant portions of the associated graded complex.

Lemma 7.2 ([Lev21, Lemma 2.8])**.** *Suppose K is a knot in S^3 such that $\epsilon(K)=-1$ and the vertical arrow from $\widetilde{\eta}_0=\widetilde{\xi}_1$ to $\widetilde{\xi}_2$ has length $k_1=1$. Then, in $\widehat{CFD}(S^3\backslash\nu(K))$, we have $D_3(\xi_2)=0$.*

7.2. **Computing $CFA^-(Q_{i,(2,2)})$.** Following the labeling of generators specified for the diagram for $Q_{i,(2,2)}$ in Figure 10, we compute the following families of discs, tabulated in Proposition 7.3. We show the shapes of these discs in Figures 11 and 12. There is a qualitative change in the nature of discs going from $i=1$ to $i=2$, but from $i=2$ to above, each family of discs simply grows linearly. We separated these discs into families, with some "special discs" that are not dependent on i, and the pure differentials which wind around the middle, which we call the "snail" discs.

Proposition 7.3. *The $CFA^-(Q_{i,(2,2)})$ will have $64i-18$ discs, giving us the following arrows:*
Families *($52i-25$ arrows)*

Source	*Arrow*	*Target*	
y_j	$U\rho_3\rho_2\rho_1$	x_j	$(0\leqslant j\leqslant 8i-4)$
y_j	$U^2\rho_3\rho_2\rho_{12}\rho_1$	x_{8i-3+j}	$(0\leqslant j\leqslant 4i-1)$
x_j	$U\rho_2\rho_1$	x_{8i-3+j}	$(0\leqslant j\leqslant 4i-1)$
y_j	ρ_1	$x_{16i-3-j}$	$(0\leqslant j\leqslant 4i-2)$
y_j	ρ_{12}	y_{8i-4-j}	$(0\leqslant j\leqslant 4i-3)$
y_j	$U\rho_{123}\rho_2\rho_1$	x_{8i-4-j}	$(0\leqslant j\leqslant 4i-3)$
y_j	$U\rho_{123}$	x_{4+j}	$(0\leqslant j\leqslant 4i-5)$
$x_{16i-3-j}$	ρ_2	y_{8i-4-j}	$(0\leqslant j\leqslant 4i-3)$
$x_{16i-3-j}$	$U\rho_{23}\rho_2\rho_1$	x_{8i-4-j}	$(0\leqslant j\leqslant 4i-3)$
$x_{16i-3-j}$	$U\rho_{23}$	x_{4+j}	$(0\leqslant j\leqslant 4i-5)$
x_{4i+j}	U	$x_{12i-3-j}$	$(0\leqslant j\leqslant 4i-2)$
y_{8i-4-j}	$U\rho_3$	x_{4+j}	$(0\leqslant j\leqslant 4i-5)$

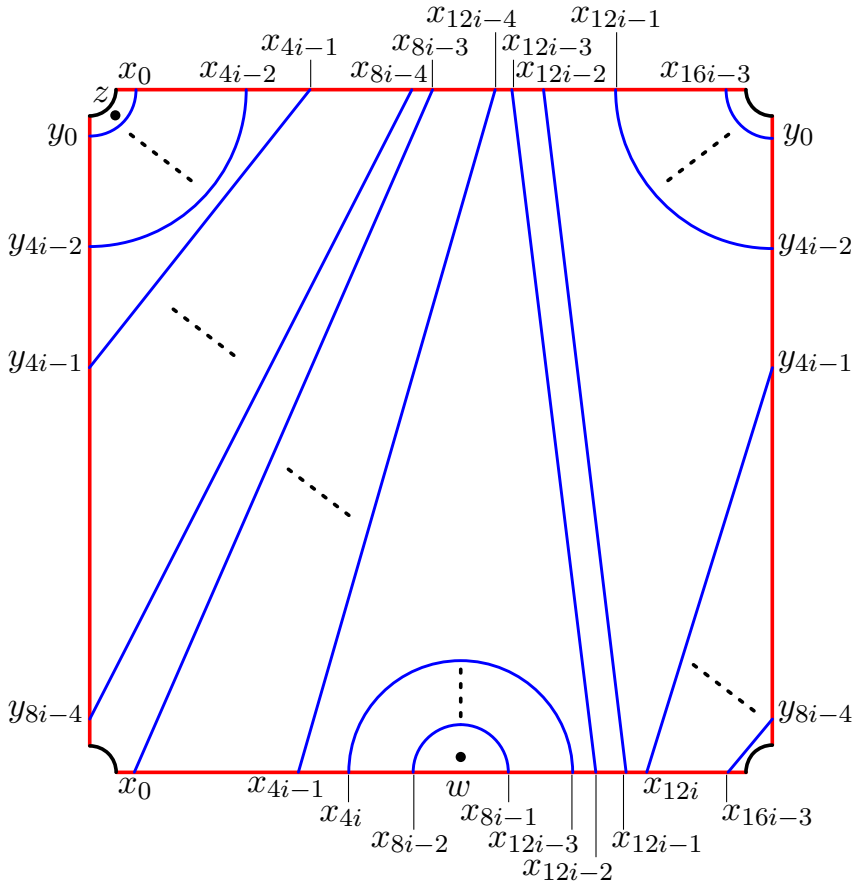


FIGURE 10. The bordered Heegaard diagram for $Q_{i,(2,2)}$ with generators labeled.

(The three families with upper bound $4i-5$ are empty when $i=1$.)

Special arrows *(14 arrows)*

Source	*Arrow*	*Target*
x_{12i+1}	U^2	x_{12i-2}
x_{12i+1}	$U^2\rho_{23}$	x_{12i-2}
y_{4i-1}	$U\rho_1$	x_{12i-2}
y_{4i-1}	$U\rho_{12}\rho_1$	x_{12i-3}
y_{4i}	$U^2\rho_1$	x_{12i-3}
y_{4i}	$U^2\rho_3$	x_{12i-2}
x_{12i}	$U^2\rho_{23}\rho_2\rho_{12}\rho_1$	x_{12i-4}
y_{4i-3}	$U^2\rho_{123}\rho_2\rho_{12}\rho_1$	x_{12i-4}
y_{4i-4}	$U^2\rho_{123}$	x_{12i-2}
y_{4i-2}	$\rho_{12}\rho_1$	x_{12i-2}
y_{4i-2}	$\rho_{12}\rho_{12}\rho_1$	x_{12i-3}
x_{12i-1}	$\rho_2\rho_1$	x_{12i-2}
x_{12i-1}	$\rho_2\rho_{12}\rho_1$	x_{12i-3}
x_{12i-2}	$\rho_2\rho_1$	x_{12i-3}

"Snail" *(all labeled U; $12i-7$ arrows)*

Source	*Arrow*	*Target*	
x_{12i}	U	x_{12i-1}	
x_{4k}	U	x_{4k+3}	$(0 \leqslant k \leqslant i-1)$
x_{4k+1}	U	x_{4k+2}	$(0 \leqslant k \leqslant i-1)$
y_{4k}	U	y_{4k+3}	$(0 \leqslant k \leqslant i-1)$
y_{4k+1}	U	y_{4k+2}	$(0 \leqslant k \leqslant i-1)$
$x_{4i+4+4k}$	U	$x_{4i+1+4k}$	$(0 \leqslant k \leqslant i-2)$
$x_{4i+3+4k}$	U	$x_{4i+2+4k}$	$(0 \leqslant k \leqslant i-2)$
$x_{8i+1+4k}$	U	$x_{8i+4+4k}$	$(0 \leqslant k \leqslant i-2)$
$x_{8i+2+4k}$	U	$x_{8i+3+4k}$	$(0 \leqslant k \leqslant i-2)$
$x_{12i+5+4k}$	U	$x_{12i+2+4k}$	$(0 \leqslant k \leqslant i-2)$
$x_{12i+4+4k}$	U	$x_{12i+3+4k}$	$(0 \leqslant k \leqslant i-2)$
$y_{4i+4+4k}$	U	$y_{4i+1+4k}$	$(0 \leqslant k \leqslant i-2)$
$y_{4i+3+4k}$	U	$y_{4i+2+4k}$	$(0 \leqslant k \leqslant i-2)$

Proof. Each of these families of discs can be shown to exist in the bordered Heegaard diagrams for $Q_{i,(2,2)}$ as shown above (using $i = 2$). Following the β curve further from each of these discs shows us that there are no other discs. □

Each arrow represents a multiplication in the A_∞ module $CFA^-(Q_{i,(2,2)})$. An arrow with source x_{12}, arrow $U^2\rho_{23}\rho_2\rho_{12}\rho_1$ and target x_8 means $m_5(x_{12}, \rho_{23}, \rho_2, \rho_{12}, \rho_1) = U^2x_8$.

7.3. **Computing the Alexander gradings of the associated graded.** To compute τ for our satellite knots, we want to leverage the following proposition:

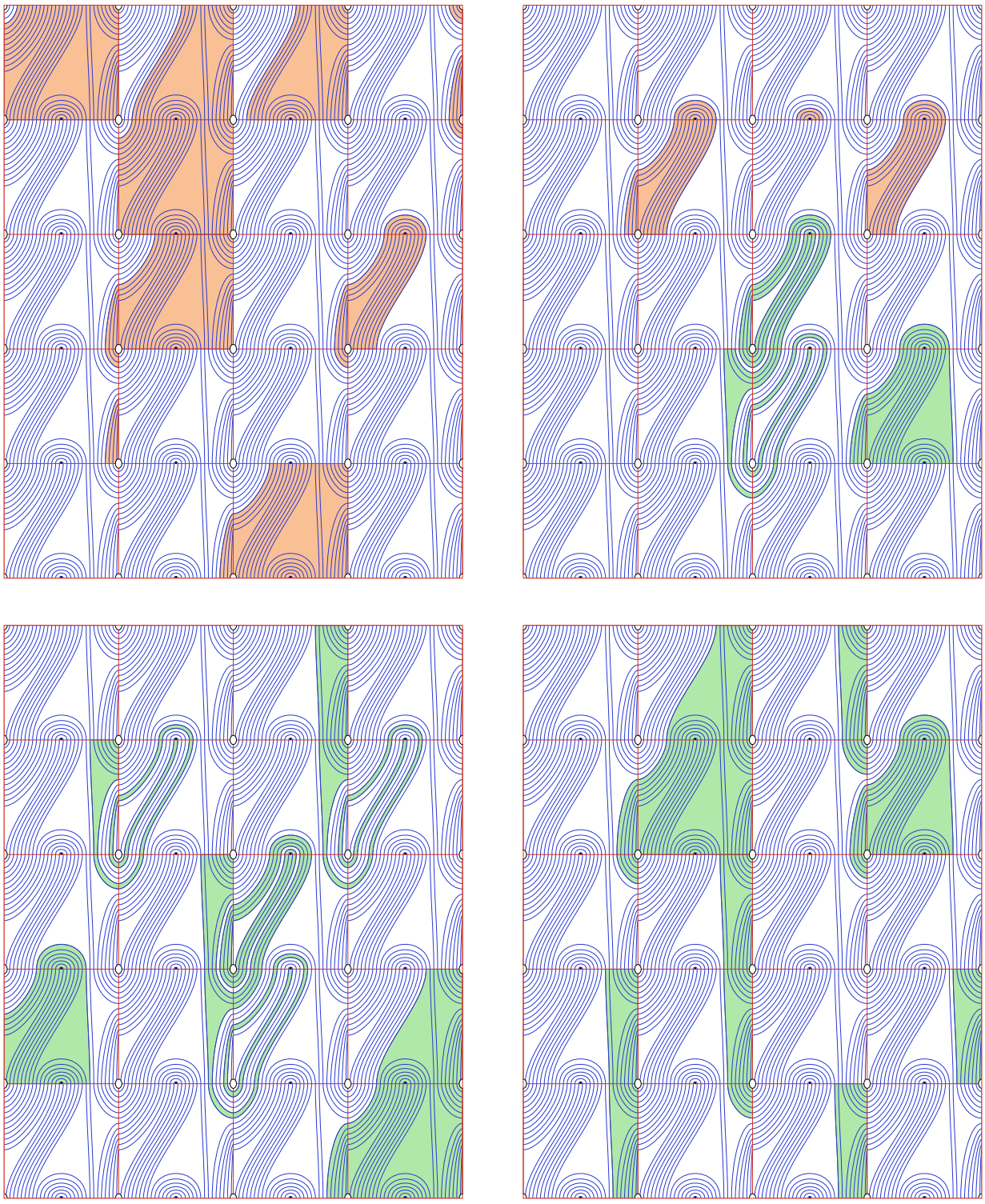


FIGURE 11. Holomorphic disks for $Q_{i,(2,2)}$: the families (orange) and the special arrows (green).

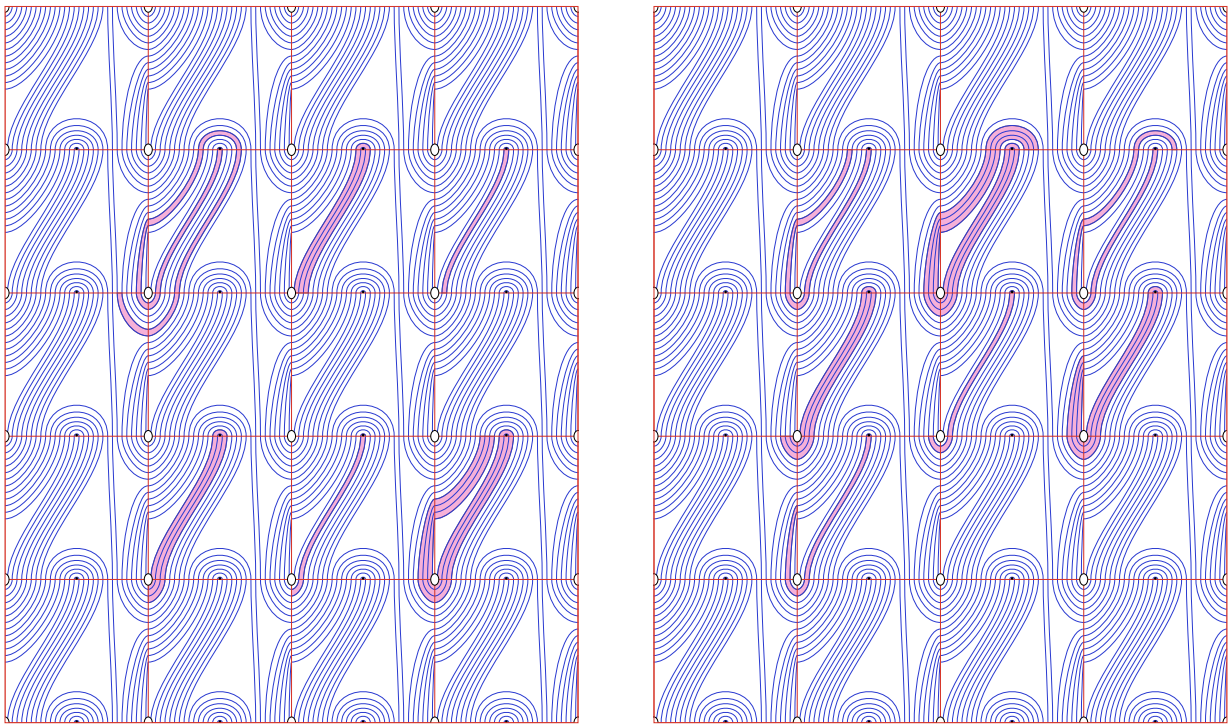


FIGURE 12. Holomorphic disks for $Q_{i,(2,2)}$, continued: the snail.

Proposition 7.4. [Lev21, Proposition 2.2] *Let $P \subset S^1 \times D^2$ be a based knot with winding number m. For each element $a \in CFA^-(S^1 \times D^2, P) \cdot \iota_0$ that is homogeneous with respect to the spin^c grading, there exists a constant c_a with the following property: For any knot $K \subset S^3$, and any homogeneous element $x \in \iota_0 \widehat{CFD}(S^3 \backslash \nu(K))$, we have*

$$A_{P(K)}(a \otimes x) \;=\; m\, A_K(x) + c_a,$$

To determine the constants c_a we compute in the simplest case, namely $K = U$ the unknot with framing 0, for which the formula reduces to:

$$A_{P(U)}(a \otimes x) \;=\; 0 \cdot A_U(x) + c_a = c_a.$$

Once we have found the desired constants, we will be able to compute τ for arbitrary companion knots in S^3, splitting cases by the signs of τ and ϵ of the companion. Since $CFK^-(U)$ is generated by a single element, $\xi_0 = \eta_0$ and by Theorem 4.6, $\widehat{CFD}(U)$ is



Thus, the box tensor product $CFA^-(Q_{i,(2,2)}) \boxtimes \widehat{CFD}(U)$ contains the $2(i-1)$ squares of the following two forms for $0 \leqslant 4j \leqslant 4i-3$:

$$\begin{array}{ccc} y_{4j} \otimes \xi_0 & \longrightarrow & y_{8i-4-4j} \otimes \xi_0 \\ {\scriptstyle U}\downarrow & & \downarrow{\scriptstyle U} \\ y_{4j+3} \otimes \xi_0 & \longrightarrow & y_{8i-7-4j} \otimes \xi_0 \end{array} \qquad \begin{array}{ccc} y_{4j+1} \otimes \xi_0 & \longrightarrow & y_{8i-5-4j} \otimes \xi_0 \\ {\scriptstyle U}\downarrow & & \downarrow{\scriptstyle U} \\ y_{4j+2} \otimes \xi_0 & \longrightarrow & y_{8i-6-4j} \otimes \xi_0 \end{array}$$

together with the following staircase, in which the two vertical U-maps can be removed by a change of basis. The change of basis is drawn in red while the blue superscripts denote the Alexander gradings of the generators.

$$\begin{array}{ccc} & & \overset{-2}{y_{4i-4}} \otimes \xi_0 \longrightarrow \overset{-2}{y_{4i}} \otimes \xi_0 + U\, y_{4i-1} \otimes \xi_0 \\ & & \not{U}\downarrow \\ \overset{-1}{y_{4i-3}} \otimes \xi_0 & \longrightarrow & \overset{-1}{y_{4i-1}} \otimes \xi_0 + U\, y_{4i-2} \otimes \xi_0 \\ \not{U}\downarrow & & \\ \overset{0}{y_{4i-2}} \otimes \xi_0 & & \end{array}$$

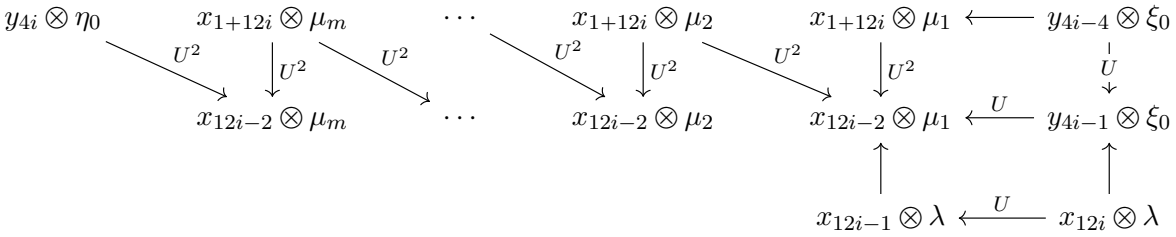
Recall that there is always guaranteed to be exactly one free $\mathbb{F}[U]$ summand. So $y_{4i-2} \otimes \xi_0$ generates the free $\mathbb{F}[U]$ summand, which gives:

$$A_{Q_{i,(2,2)}(U)}(y_{4i-2} \otimes \xi_0) = 0,\ A_{Q_{i,(2,2)}(U)}(y_{4i-3} \otimes \xi_0) = -1,\ \ldots,\ A_{Q_{i,(2,2)}(U)}(y_{4i} \otimes \xi_0) = -2.$$

Therefore, we have that $c_{y_{4i-2}} = 0,\ c_{y_{4i-3}} = c_{y_{4i-1}} = -1,\ c_{y_{4i}} = c_{y_{4i-4}} = -2$.

7.4. **Computation of $gCFK(Q_{i,(2,2)}(K))$ and $\tau(Q_{i,(2,2)}(K))$.** Now, we must consider the tensor product $CFA^-(Q_{i,(2,2)}) \boxtimes \widehat{CFD}(S^3 \backslash \nu(K))$ appearing in the pairing theorem. It suffices to consider the unstable neighborhood, as mentioned above. Indeed, we consider the unstable chains specified by Theorem 4.6, and tensor with the CFA^- structure described in the previous section.

- $\epsilon(K) = 0$: As documented in [Lev21] this case reduces to that of the unknot, and thus $\tau(Q_{i,(2,2)}(K)) = -A_{Q_{i,(2,2)}(U)}(y_{4i-2} \otimes \xi_0) = 0$.
- $\tau(K) > 0$ and $\epsilon(K) = 1$: Our isolated summand of the $gCFK$ becomes:



The free $\mathbb{F}[U]$ summand generator for $gCFK(Q_{i,(2,2)}(K))$ is given by

$$g_+ = y_{4i} \otimes \eta_0 + x_{1+12i} \otimes \mu_m + \cdots + x_{1+12i} \otimes \mu_2 + x_{1+12i} \otimes \mu_1.$$

To see that this is a cycle, we compute

$$\partial^{\boxtimes}(g_+) = 2U^2 x_{12i-2} \otimes \mu_m + \cdots + 2U^2 x_{12i-2} \otimes \mu_1 \equiv 0 \pmod 2.$$

Now, notice that no U power multiple of $y_{4i} \otimes \eta_0$ is in the image of any generator in the summand, so that $g_+ \notin \mathrm{im}\partial^{\boxtimes}$. Hence, free $\mathbb{F}[U]$ summand is indeed generated by g_+, and using the general facts

$$A(\partial a) = A(a) \quad \text{and} \quad A(Ua) = A(a) - 1,$$

along with

$$A_{Q_{i,(2,2)}}(y_{4i} \otimes \eta_0) = mA_K(\eta_0) + C_{y_{4i}} = -2,$$

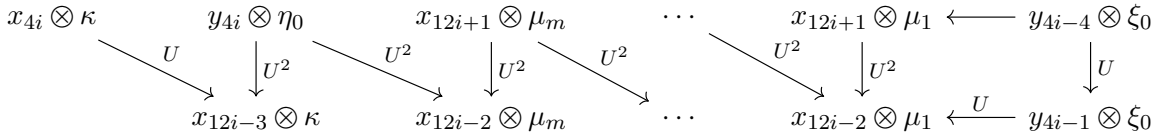
we obtain

$$A_{Q_{i,(2,2)}}(x_{12i+1} \otimes \mu_i) = 0, \quad A_{Q_{i,(2,2)}}(x_{12i-2} \otimes \mu_i) = -2, \quad (i = 1, \ldots, m).$$

Thus,

$$\tau(Q_{i,(2,2)}(K)) = -A_{Q_{i,(2,2)}}(g_+) = 2.$$

- $\tau(K) > 0$ and $\epsilon(K) = -1$:



We know that

$$A_{Q_{i,(2,2)}(K)}(y_{4i} \otimes \eta_0) = mA_K(\eta_0) + C_{y_{4i}} = -2,$$

so we obtain

$$A_{Q_{i,(2,2)}(K)}(x_{12i+1} \otimes \mu_i) = 0, \quad A_{Q_{i,(2,2)}(K)}(x_{12i-2} \otimes \mu_i) = -2, \quad (i = 1, \ldots, m).$$

The free $\mathbb{F}[U]$ summand generator for $gCFK(Q_{i,(2,2)}(K))$ is then given by

$$g_- = Ux_{4i} \otimes \kappa + y_{4i} \otimes \eta_0 + x_{12i+1} \otimes \mu_m + \cdots + x_{12i+1} \otimes \mu_2 + x_{12i+1} \otimes \mu_1.$$

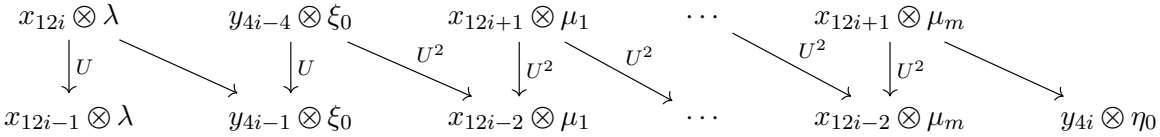
To see this, notice that

$$\partial^{\boxtimes}(g_-) = 2U^2 x_{12i-3} \otimes \kappa + 2x_{\Delta k_q 2i-2} \otimes \mu_m + \cdots + 2x_{12i-2} \otimes \mu_2 + 2x_{12i-2} \otimes \mu_1 \equiv 0 \pmod 2,$$

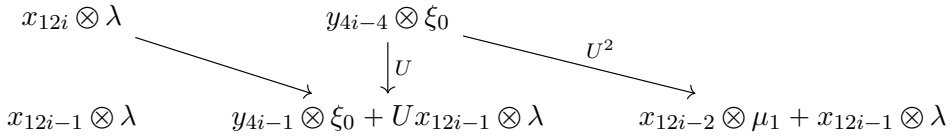
so that g_- is in the kernel of $\partial^{\boxtimes}$, yet note that no U power multiple of $y_{4i} \otimes \eta_0$ is in the image of $\partial^{\boxtimes}$, so in particular, the sum given by g_- is indeed the free $\mathbb{F}[U]$ generator. Hence,

$$\tau(Q_{i,(2,2)}(K)) = -A_{Q_{i,(2,2)}(K)}(g_-) = 2.$$

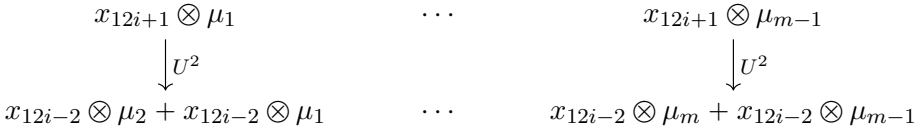
- $\tau(K) < 0$ and $\epsilon(K) = 1$: Our tensor structure becomes

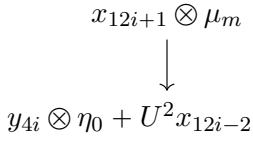


After a change of basis, we obtain:



along with summands of the form





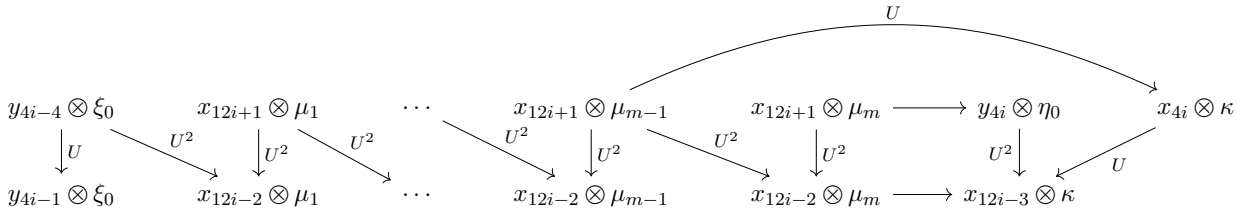
Hence, we conclude that $x_{12i-1} \otimes \lambda$ is the generator of the free $\mathbb{F}[U]$ summand in our $gCFK$. Now, from the fact that

$$A_{Q_{i,(2,2)}(K)}(y_{4i-1} \otimes \xi_0) = -1,$$

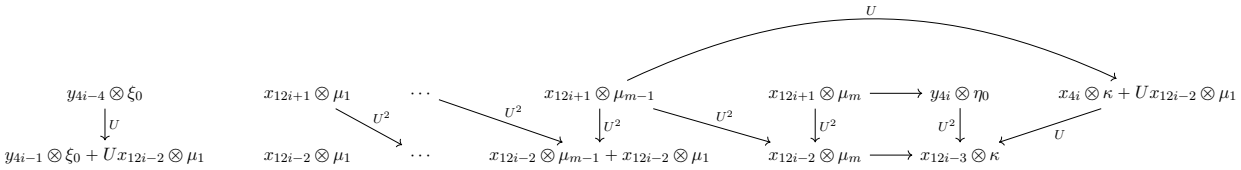
we obtain that

$$\tau(Q_{i,(2,2)}(K)) = -A_{Q_{i,(2,2)}(K)}(x_{12i-1} \otimes \lambda) = 0.$$

- $\tau(K) < 0$ and $\epsilon(K) = -1$:



After a change of basis, we obtain the following summand:



where we send $x_{12i-2}\otimes\mu_i \mapsto x_{12i-2}\otimes\mu_i + x_{12i-2}\otimes\mu_1$ for $i = 2,\dots,m-1$.

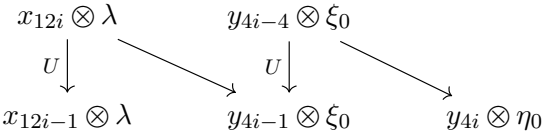
Notice then that $x_{12i-2}\otimes\mu_1$ must be the generator of the free $\mathbb{F}[U]$ summand in our $gCFK$. Then, we see that:

$$A_{Q_{i,(2,2)}}(y_{4i}\otimes\eta_0) = -2,$$

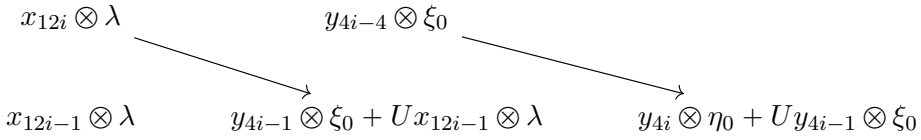
so using our formulas relating the differential and U to the Alexander grading, we obtain

$$\tau(Q_{i,(2,2)}(K)) = -A_{Q_{i,(2,2)}}(x_{12i-2}\otimes\mu_1) = 0.$$

- $\tau(K) = 0$ and $\epsilon(K) = 1$:



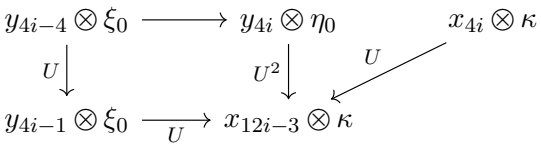
After changing basis, we obtain



As above, our generator for the free $\mathbb{F}[U]$ summand will be $x_{12i-1}\otimes\lambda$, and since $y_{4i}\otimes\eta_0$ has Alexander grading -2, we see that

$$\tau(Q_{i,(2,2)}(K)) = -A_{Q_{i,(2,2)}}(x_{12i-1}\otimes\lambda) = 0.$$

- $\tau(K) = 0$ and $\epsilon(K) = -1$:



In this case, the generator of the free summand becomes $g = y_{4i-1}\otimes\xi_0 + x_{4i}\otimes\kappa$: none of its U-powers are in the image of $\partial^{\boxtimes}$ since no U-powers of $x_{4i}\otimes\kappa$ are, and yet $\partial^{\boxtimes}(g) = 2x_{12i-3}\otimes\kappa = 0$. Note that

$$A_{Q_{i,(2,2)}}(y_{4i}\otimes\eta_0) = -2,$$

so that

$$\tau(Q_{i,(2,2)}(K)) = -A_{Q_{i,(2,2)}}(g) = 1.$$

Hence, we obtain

$$\tau(Q_{i,(2,2)}(K)) = \begin{cases} 0 & \text{if } \tau(K) < 0, \text{ or } \tau(K) = 0 \text{ and } \epsilon(K) = 0 \text{ or } 1, \\ 1 & \text{if } \tau(K) = 0 \text{ and } \epsilon(K) = -1, \\ 2 & \text{if } \tau(K) > 0, \end{cases}$$

which matches the $(m, n) = (2, 2)$ case in Theorem 1.5 in [PX24].

References

[Bod24] Holt Bodish. Genus, fiberedness, τ and ϵ of satellite knots with n-twisted generalized Mazur patterns, 2024. [2], [3]

[Che23] Wenzhao Chen. Knot floer homology of satellite knots with (1, 1) patterns: W. chen. *Selecta Mathematica*, 29(4):53, 2023. [3], [17], [19], [21], [22], [24]

[Con70] J. H. Conway. An enumeration of knots and links, and some of their algebraic properties. In John Leech, editor, *Computational Problems in Abstract Algebra (Proc. Conf., Oxford, 1967)*, pages 329–358. Pergamon, Oxford, 1970. [21]

[Hom14] Jennifer Hom. Bordered heegaard floer homology and the tau-invariant of cable knots. *Journal of Topology*, 7(2):287–326, 2014. [1], [8], [10], [11], [12], [13]

[Hom17] Jennifer Hom. A survey on heegaard floer homology and concordance. *Journal of Knot Theory and Its Ramifications*, 26(02):1740015, 2017. [9], [11], [12]

[Hom20] Jennifer Hom. Lectures notes on heegaard floer homology, 2020. [11]

[HPC21] Matthew Hedden and Juanita Pinzón-Caicedo. Satellites of infinite rank in the smooth concordance group. *Inventiones mathematicae*, 225(1):131–157, January 2021. [1]

[HRW23] Jonathan Hanselman, Jacob Rasmussen, and Liam Watson. Bordered floer homology for manifolds with torus boundary via immersed curves, 2023. [2]

[Kir97] Robion C. Kirby. Problems in low-dimensional topology. In *Geometric Topology (Athens, GA, 1993)*, volume 2 of *AMS/IP Studies in Advanced Mathematics*, pages 35–473. American Mathematical Society, Providence, RI, 1997. [3]

[Lev21] Adam Simon Levine. Non-surjective satellite operators and piecewise-linear concordance, 2021. [1], [11], [12], [18], [27], [31], [32]

[LOT18] Robert Lipshitz, Peter Ozsváth, and Dylan Thurston. *Bordered Heegaard Floer homology: Invariance and pairing*. Memoirs of the American Mathematical Society, 2018. [13], [16], [18]

[Man16] Ciprian Manolescu. An introduction to knot floer homology, 2016. [8]

[Moi52] Edwin E. Moise. Affine structures in 3-manifolds. V. The triangulation theorem and Hauptvermutung. *Ann. of Math. (2)*, 56:96–114, 1952. [5]

[OS03] Peter Ozsváth and Zoltán Szabó. Knot Floer homology and the four-ball genus. *Geometry & Topology*, 7:615–639, 2003. [1]

[OS04a] Peter Ozsváth and Zoltán Szabó. Holomorphic discs and genus bounds. *Geom. Topol.*, 8:311–334, 2004. [11]

[OS04b] Peter Ozsváth and Zoltán Szabó. Holomorphic discs and knot invariants. *Advances in Mathematics*, 186(1):58–116, 2004. [8], [11], [12]

[OS04c] Peter Ozsváth and Zoltán Szabó. Holomorphic discs and three-manifold invariants: properties and applications. *Annals of Mathematics (2)*, 159(3):1159–1245, 2004. arXiv:math/0105202. [4], [7]

[OS04d] Peter Ozsváth and Zoltán Szabó. Holomorphic discs and topological invariants for closed three-manifolds. *Annals of Mathematics (2)*, 159(3):1027–1158, 2004. arXiv:math/0101206. [4], [8]

[OS06] Peter Ozsváth and Zoltan Szabó. An introduction to heegaard floer homology, 2006. [5], [6], [7], [8]

[OS17] Peter Ozsváth and Zoltan Szabó. An overview of knot floer homology, 2017. [9]

[OST08] Peter S. Ozsváth, Zoltán Szabó, and Dylan Thurston. Legendrian knots, transverse knots and combinatorial Floer homology. *Geom. Topol.*, 12(2):941–980, 2008. [11]

[PX24] Jay Patwardhan and Zheheng Xiao. Generalized Mazur patterns and immersed Heegaard Floer homology, 2024. [2], [3], [21], [36]

[Ras03] Jacob A. Rasmussen. *Floer homology and knot complements*. PhD thesis, Harvard University, 2003. arXiv:math/0306378. [8], [11], [12]

[RS93] Joel Robbin and Dietmar Salamon. The Maslov index for paths. *Topology*, 32(4):827–844, 1993. [7]

[Sch56] Horst Schubert. Knoten mit zwei Brücken. *Math. Z.*, 65:133–170, 1956. [19], [21]

[Sei35] H. Seifert. Über das geschlecht von knoten. *Math. Ann.*, 110(1):571–592, 1935. [11]

Department of Mathematics, Dartmouth College, Hanover, NH 03755
Email address: ribhu.hooja.27@dartmouth.edu

Department of Mathematics, Dartmouth College, Hanover, NH 03755
Email address: sean.kim.29@dartmouth.edu

Department of Mathematics, Dartmouth College, Hanover, NH 03755
Email address: frupchin@gmail.com

Department of Mathematics, Dartmouth College, Hanover, NH 03755
Email address: shreya.sinha.gr@dartmouth.edu

Department of Mathematics, Dartmouth College, Hanover, NH 03755
Email address: allison.l.tsypin.gr@dartmouth.edu